

Mathematical Control Theory and the challenges of Systems Molecular Biology

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Objectives of Talk

- *review some questions in “systems” molecular biology*
- *show how new control/systems theoretical problems arise*
- *speculate on potential impact on molecular biology*

Outline

■ *introduction: biochemistry*

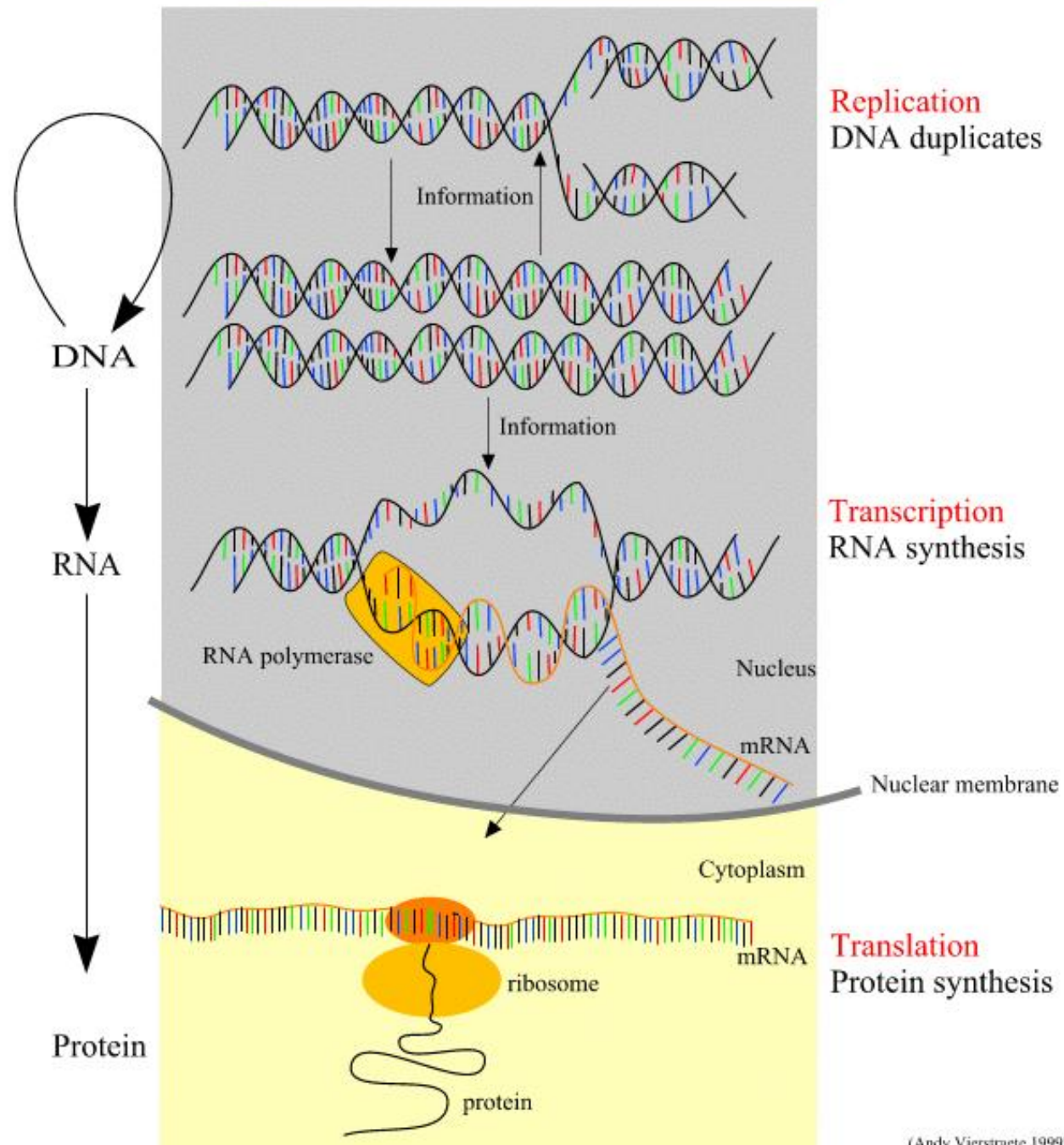
- central dogma
- life $\sim$ protein [and gene] networks
- mathematical modeling - why now?
- measurement technologies
- (I/O)systems-theoretic questions

■ *focus: “adaptation” & “robustness”*

- inhibitory feedback in gene regulation
- integral control in chemotaxis
- robust stability structures; kinetic proofreading
- interconnections, small gain, & MAPK cascades
- measurement noise & hybrid systems

■ *discussion*

The Central Dogma of Molecular Biology



not quite right ... retroviruses, reverse transcription $\text{DNA} \leftarrow \text{RNA}$; also RNA viruses $\text{RNA} \leftrightarrow \text{RNA}$

Genomics

DNA encoded in 4-letter alphabet {A,T,C,G}

```
gcacgagtaa acatgcactt cccaggccac agcagcaaga aggaggaatc tgcccaagcg  
gccctcacga agctgaactc ctgggtcccc accaccaaga accccgtcat catcagcgcc  
cccatgtatc tcatcgccaa cggcactctt gcggccgagg tatccaaggc cggcgggtatt  
ggctttgtcg ccggcggctc cgacttccgc cccgggtcct cccacctaac cgccctctct  
accgaactcg cctccgcccg cagccgcctc ggtcttaccg accgccccct caccctctct  
cccggcattg gcgtcggcct cattttaacc cacaccatct ccgttcccta cgtaaccgac
```

DNA → RNA is basically 1-1; just replace T by U

RNA → proteins: via genetic code (modulo some “editing”)
obtain complete “parts list” for organism

Genetic Code

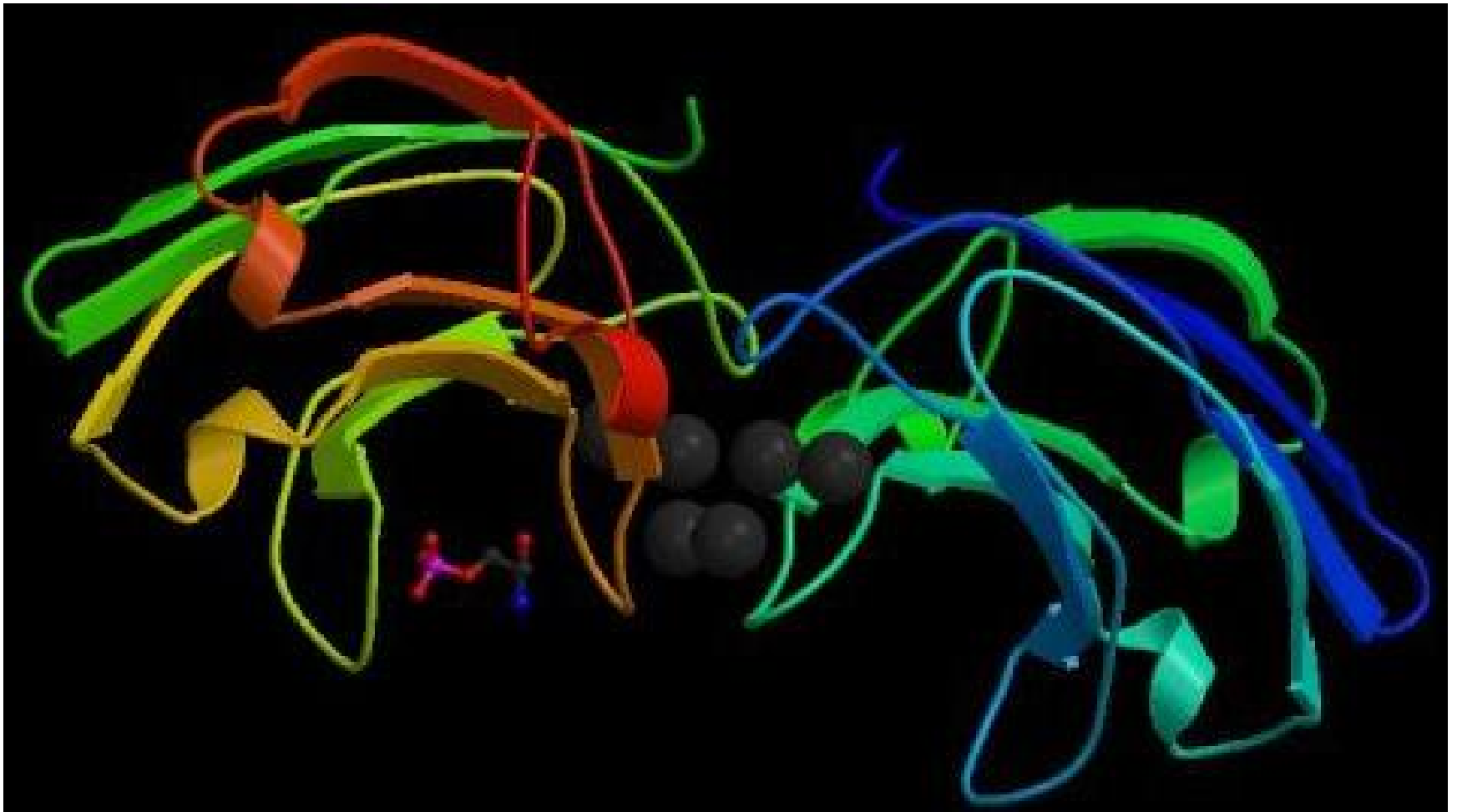
$\Sigma_n = \{U, A, G, C\}$, $\Sigma_a = \{A, R, D, N, C, E, Q, G, H, I, L, K, M, F, P, S, T, W, Y, V\}$
 homomorphism $\Sigma_n^* \rightarrow \Sigma_a^*$ (nucleotides, aminoacids)
 which extends $\Sigma_n^3 \rightarrow \Sigma_a$ given by:

UUU UUC UUA UUG	phenyl alanine leucine	UCU UCC UCA UCG	serine	UAU UAC UAA UAG	tyrosine stop	UGU UGC UGA UGG	cysteine stop tryptophan
CUU CUC CUA CUG	leucine	CCU CCC CCA CCG	proline	CAU CAC CAA CAG	histidine glutamine	CGU CGC CGA CGG	arginine
AUU AUC AUA AUG	isoleucine methionine	ACU ACC ACA ACG	threonine	AAU AAC AAA AAG	asparagine lysine	AGU AGC AGA AGG	serine arginine
GUU GUC GUA GUG	valine	GCU GCC GCA GCG	alanine	GAU GAC GAA GAG	aspartic acid glutamic acid	GGU GGC GGA GGG	glycine

A=adenine G=guanine C=cytosine U=uracil [$\leftrightarrow$ T=thymine]
 e.g. TACTCATTGCGC $\rightarrow$ AUGAGUAACGCG $\rightarrow$ MetSerAsnAla

Proteomics

three-dimensional shape of protein $\Rightarrow$ function



(lego-like fitting, enzymatic capabilities, depend on shape)

Proteins

primary components of living things

- enzymes making life's chemical reactions possible
- switches that control whether genes on/off
- sensors to “touch, smell, taste, ...”
- actuators that make muscles move (myosin, actin)
- detectors for immune response (self/other)
- structural support that holds creature together

*& they direct the breakdown/reassembly
of the others: lipids, sugars, ...*

Is the Blueprint Enough?

since (basically) genome $\Rightarrow$ proteome,

natural conjecture:

$$\begin{array}{c} \textit{(almost) genetic identity} \\ \Rightarrow \\ \textit{(almost) equal organism} \end{array}$$

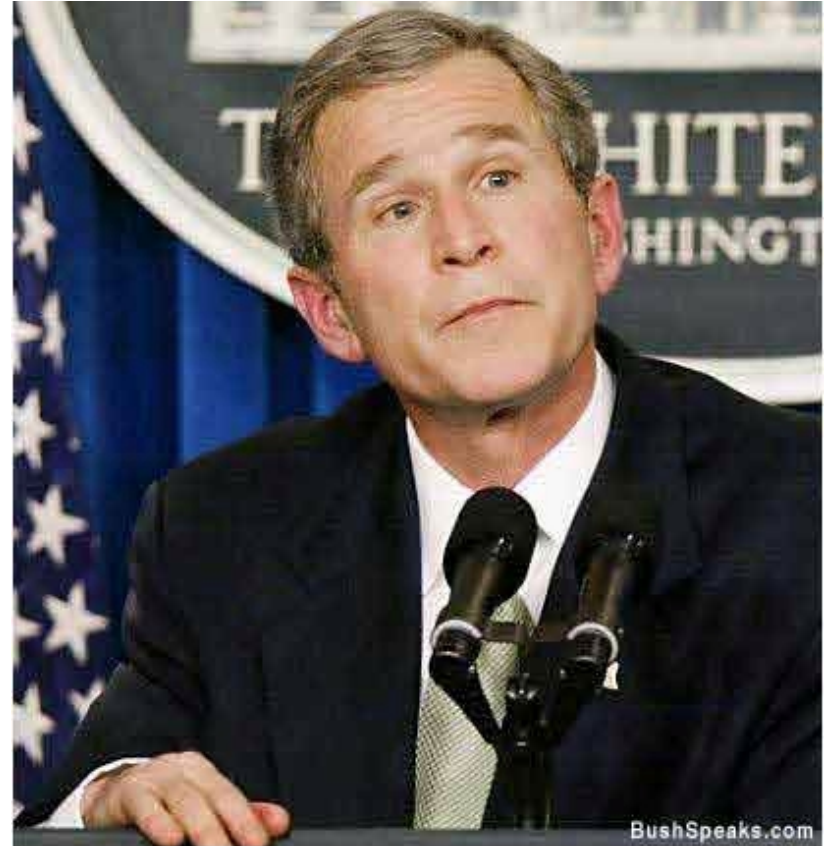
if this were true, then we'd have the following corollary:

(applying Wolfgang Enard et al, Science, April 2002)

Corollary



|||



(98.7% identical DNA sequences)

obviously a contradiction: different intelligence, etc
what gives?

Proteome to “Physiome”

proteins interact (enzymatic reaction networks)

and feed-back to control gene expression
(at top level of “central dogma”)

— *which* genes expressed (active) *where* and *when*?

feedback loops introduce dynamics

one $\neq$ feedback loop, or change in a reaction parameter,
may well result in totally different steady states
(multi-stability as a mechanism for development)

and small $\neq$ in initial states may $\Rightarrow$ huge $\neq$ trajectories
(sensitive dependence)

*development and behavior are characteristics
of a highly interconnected dynamical system*

Science

magazine

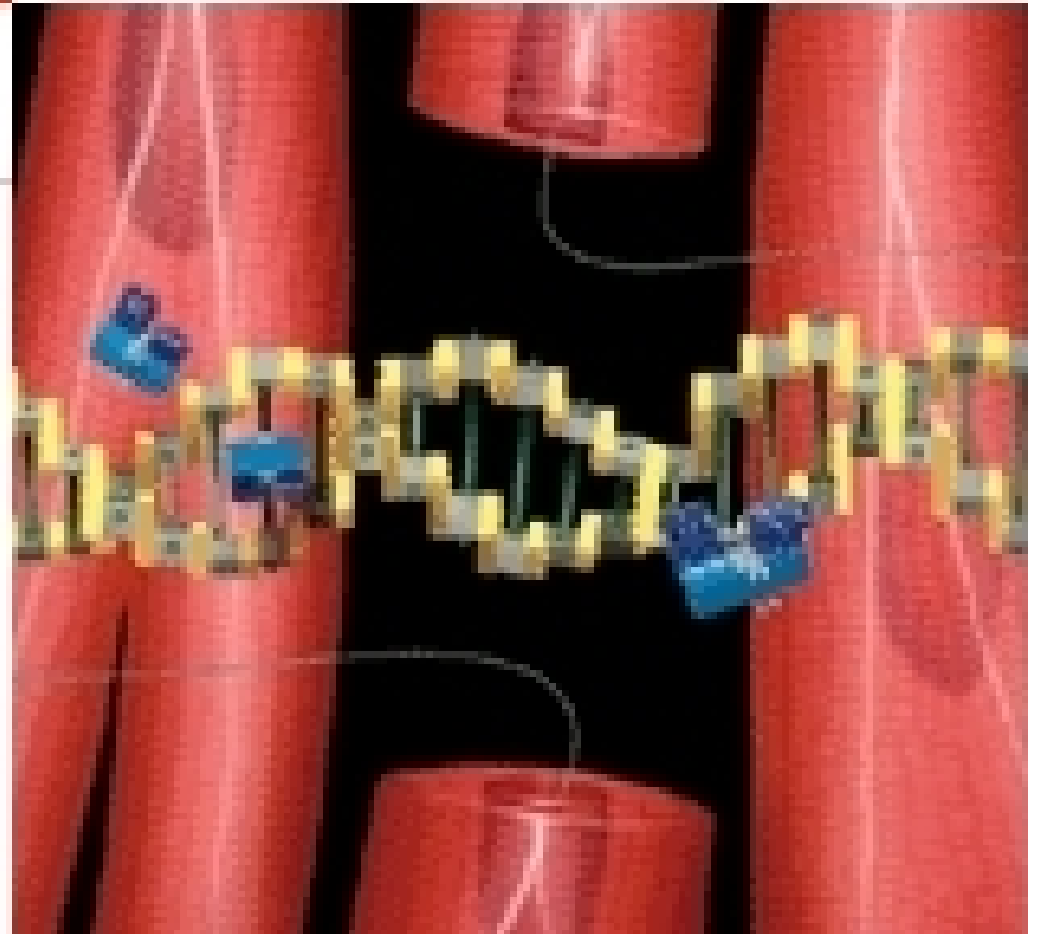
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Volume 295

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Systems Biology



“Systems” Biology

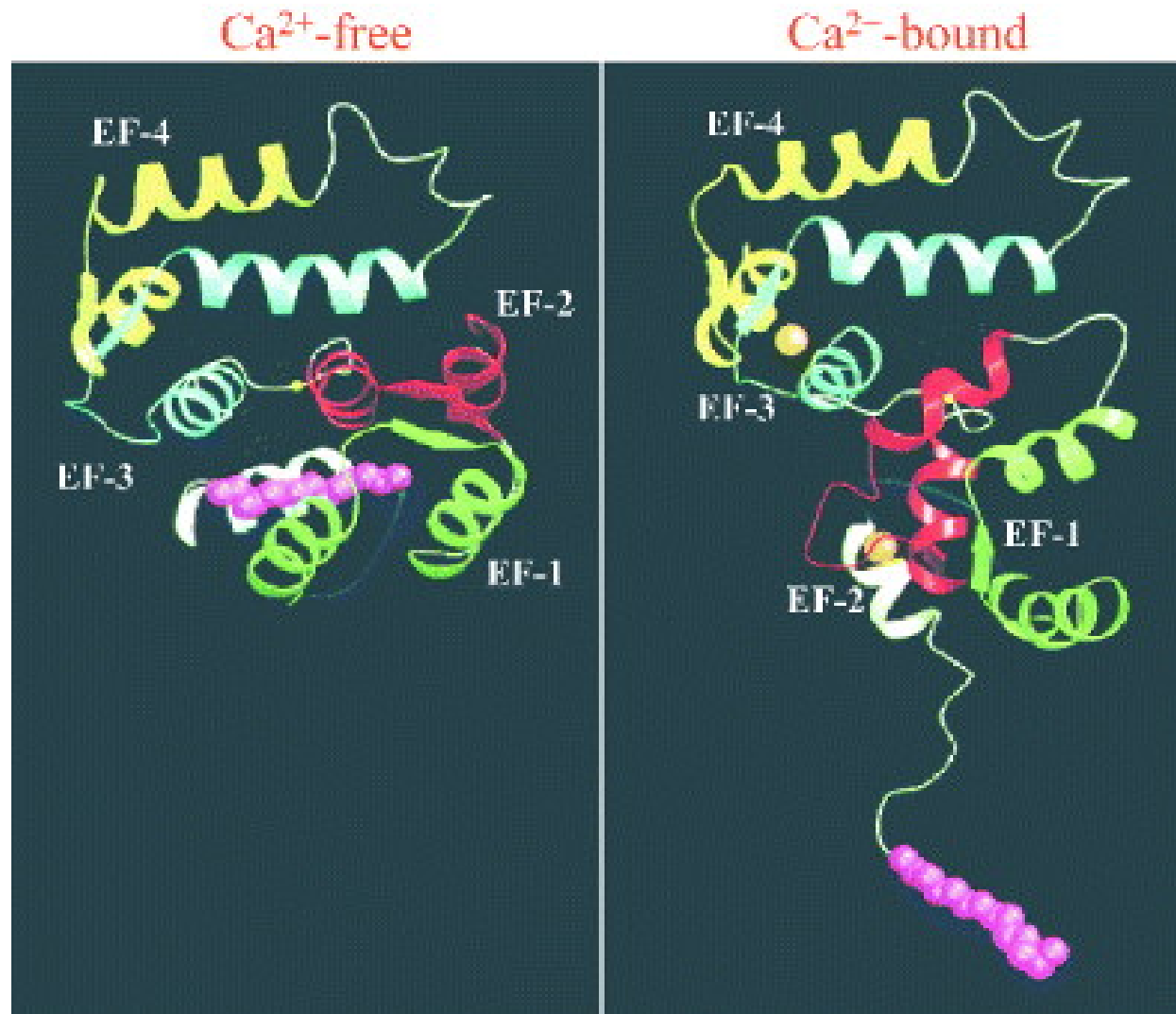
cell life based on interactions among proteins
and their regulation of gene expression through (e.g.)

- “lock and key” binding
- transfer of small molecules

e.g. phosphorylation reaction: a kinase (enzyme) transfers
phosphate group (PO_4) from ATP molecule $\rightsquigarrow$ target protein
phosphate group causes structural (conformational) change
in target protein (“click” into new position),
which will then bind to, or release, some other molecule

huge “wireless network” of such interactions
driven by external physical (UV radiation, temperature)
and chemical signals (drugs, hormones, nutrients)

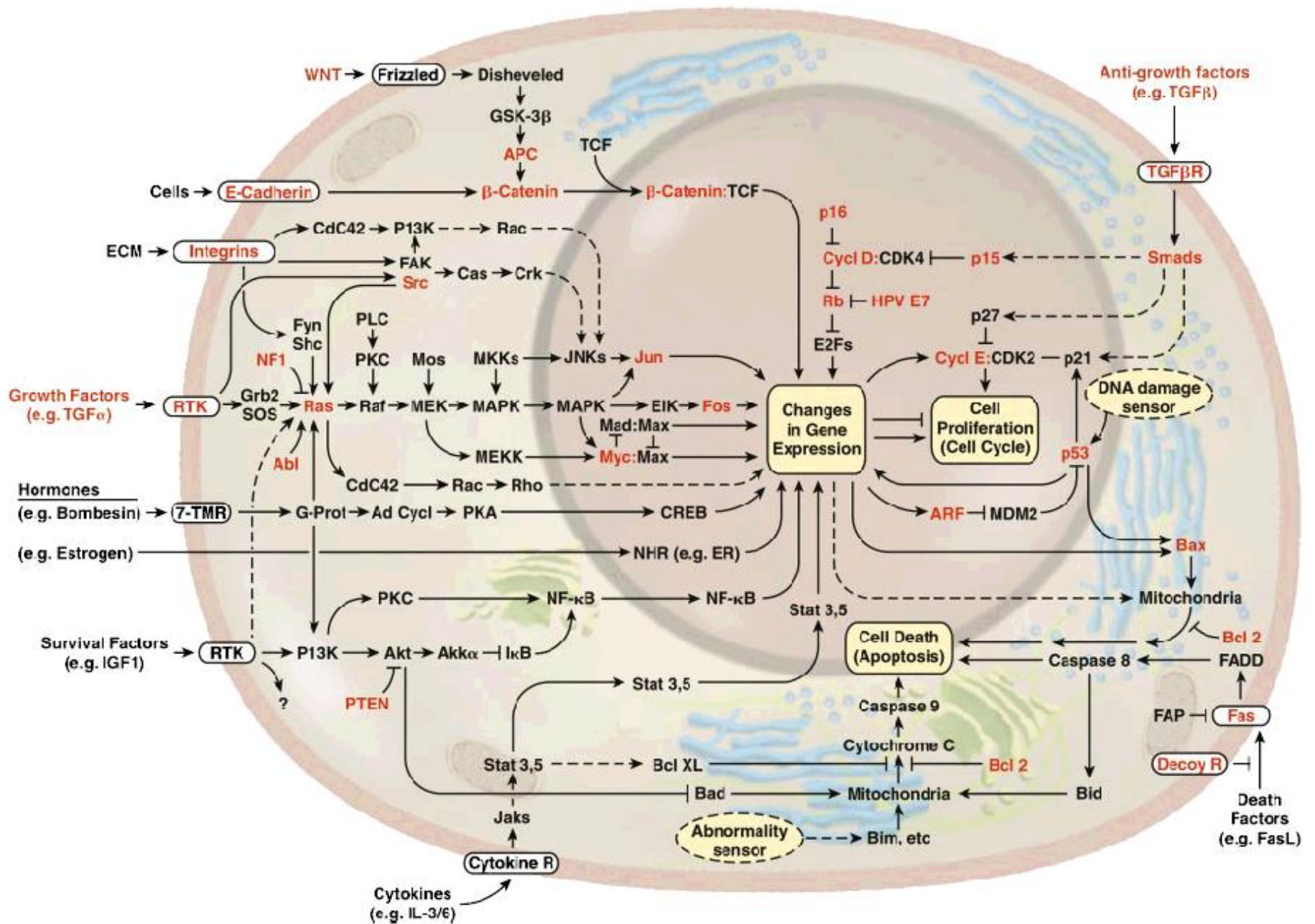
Conformation Changes



Ca^{2+} binds (at two sites) to recoverin, inducing shape change

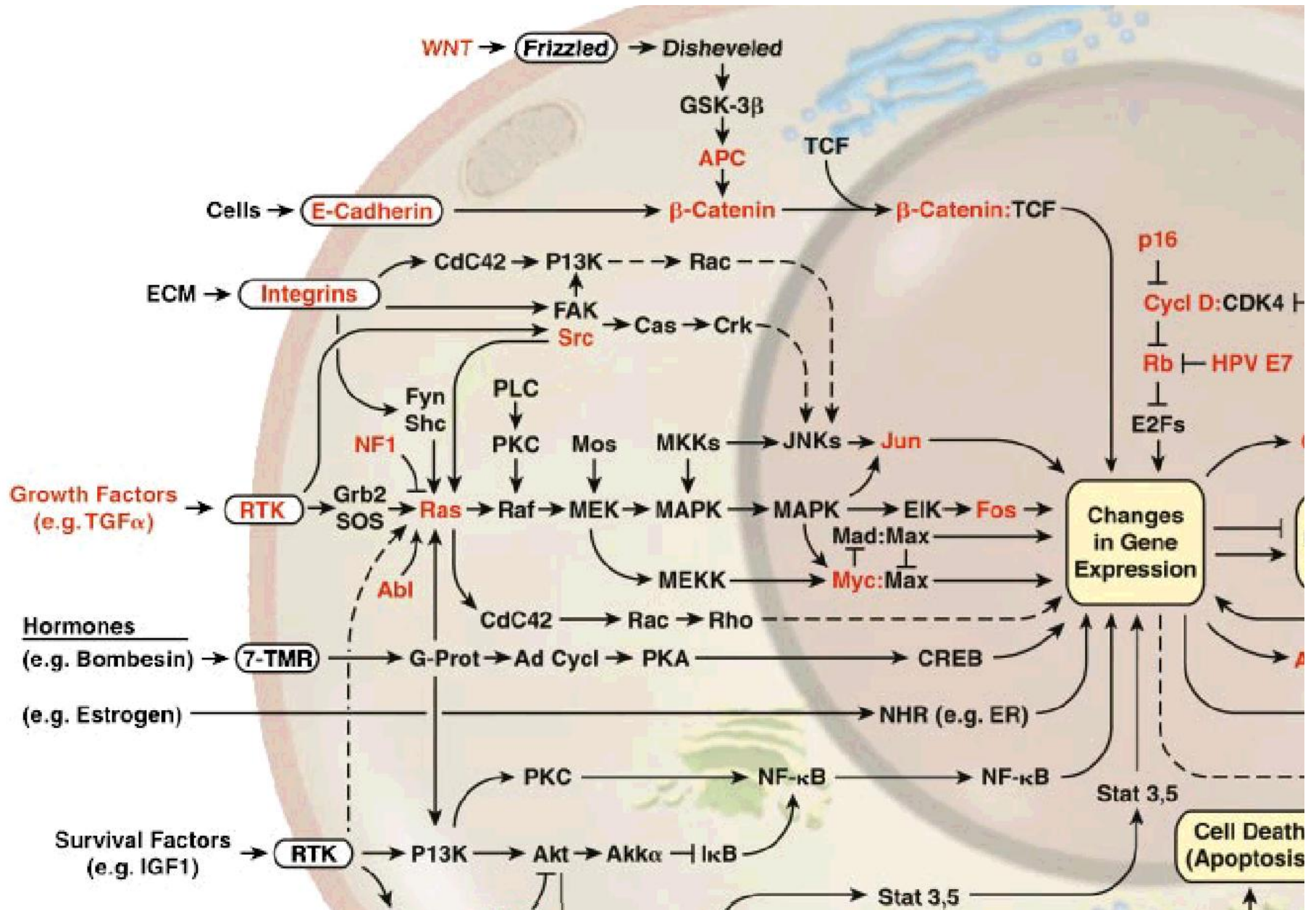
recoverin is a protein that acts as calcium sensor in retinal rod cells;
it controls the lifetime of photoexcited rhodopsin by inhibiting rhodopsin kinase <http://www.biochemj.org/>

E.g.: Cancer Network

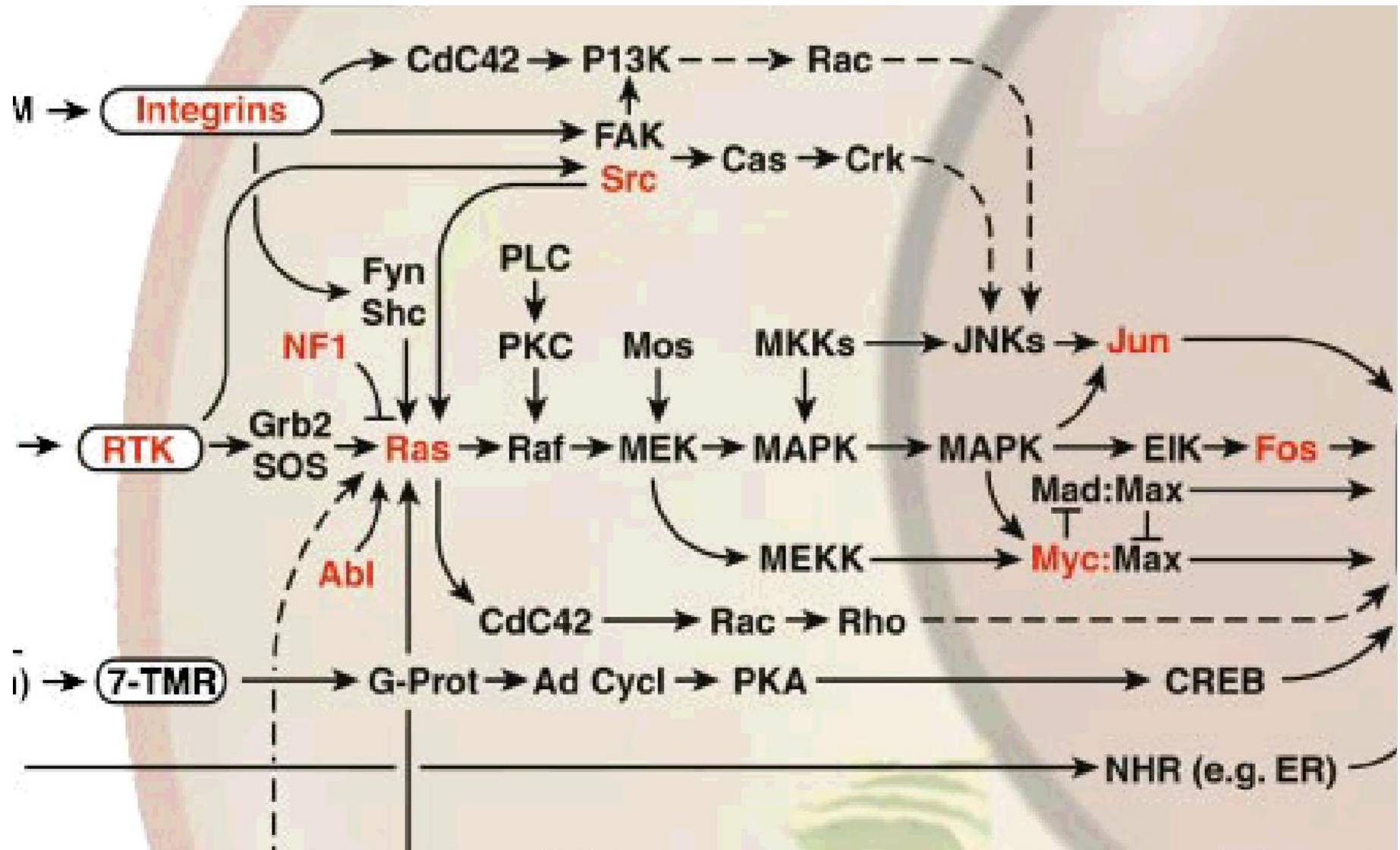


The hallmarks of cancer, Hanahan & Weinberg, Cell 2000

Zoom-in



Zoom more



Signaling Cascade Animation

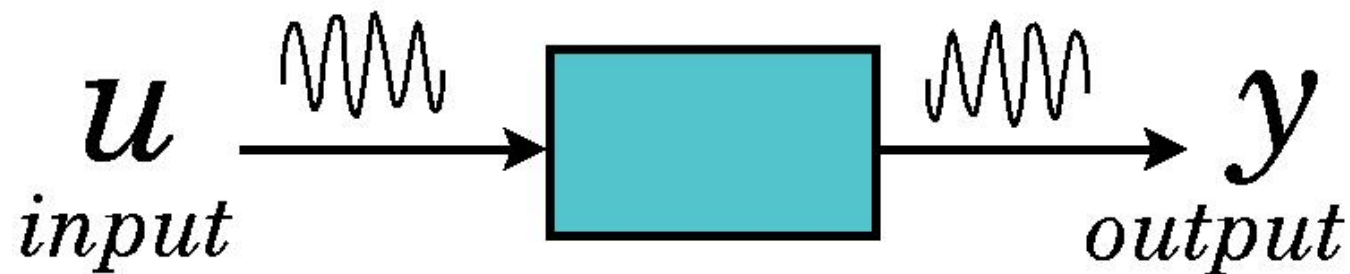
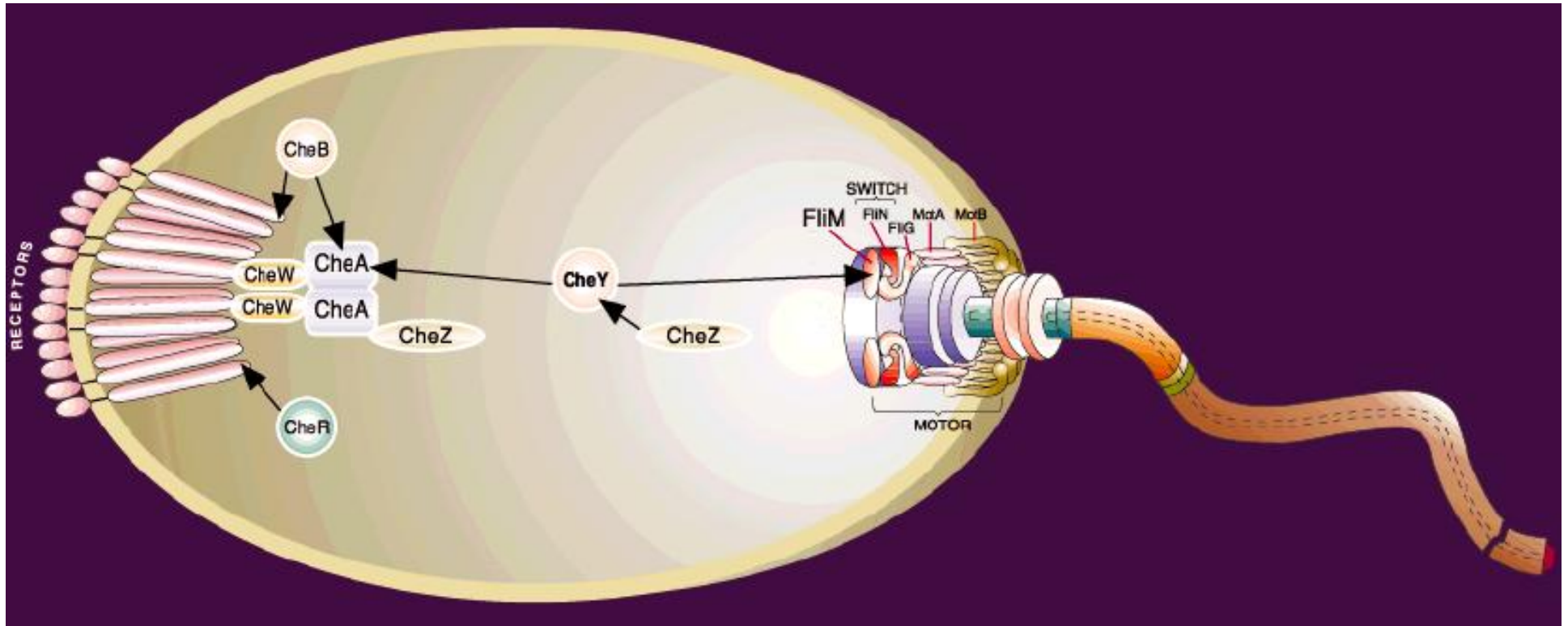
MAPK animation

(from biocreations.com)

(back to signaling cascades later)

another example of intracellular signaling system:

E.coli Chemotactic Subsystem



a complete I/O system: sensors, states, outputs (more later)

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- **mathematical modeling - why now?**
- measurement technologies
- (I/O)systems-theoretic questions

■ *focus: “adaptation” & “robustness”*

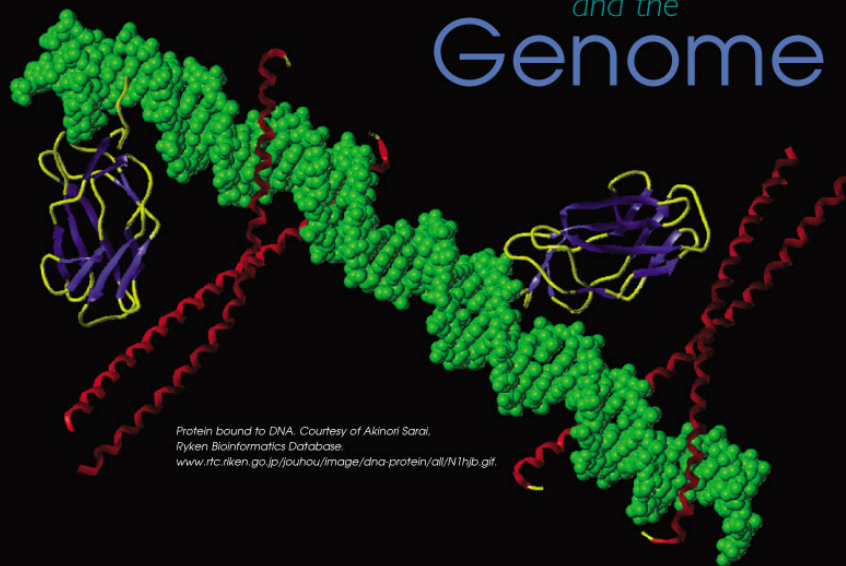
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Mathematics

and the

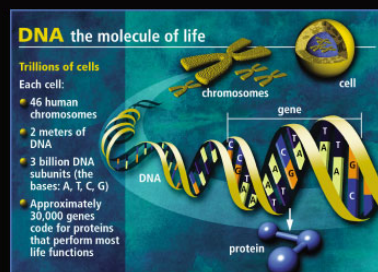
Genome



Protein bound to DNA. Courtesy of Akinori Sarai, Riken Bioinformatics Database. www.rtc.riken.go.jp/jouhou/image/dna-protein/all/N1hjb.gif.

Mathematics makes possible the management and analysis of the Human Genome Project's massive database. Numerical analysis, statistics, and modeling play a significant role in mapping and sequencing our DNA—the blueprint for the genetic information that determines what makes each of us unique. Researchers predict that this fusion of mathematics and biology will result in a new era of molecular medicine, when the diagnosis, treatment, and prevention of disease will be individual-specific and thus more successful.

www.mathforum.org/mam/02/



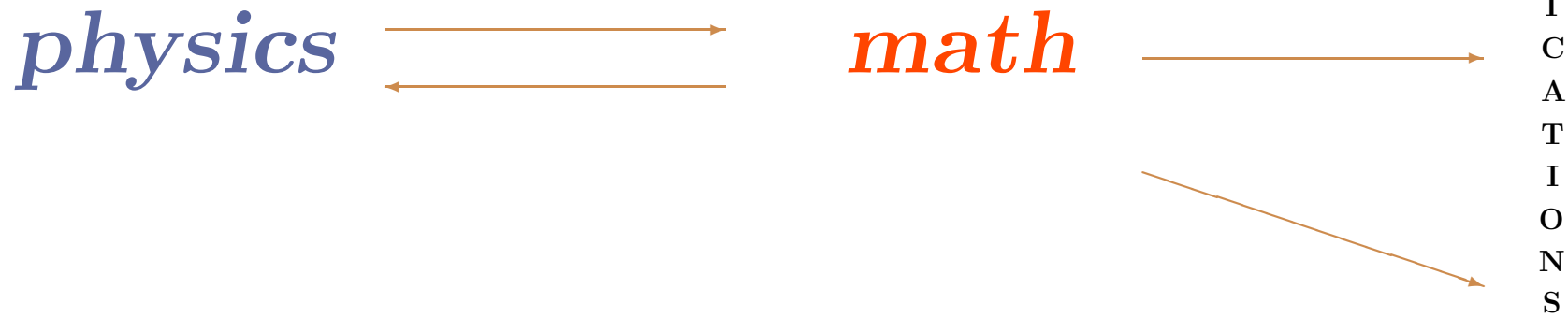
Courtesy of U.S. Department of Energy Human Genome Program. www.ornl.gov/hgmis.

Mathematics Awareness Month

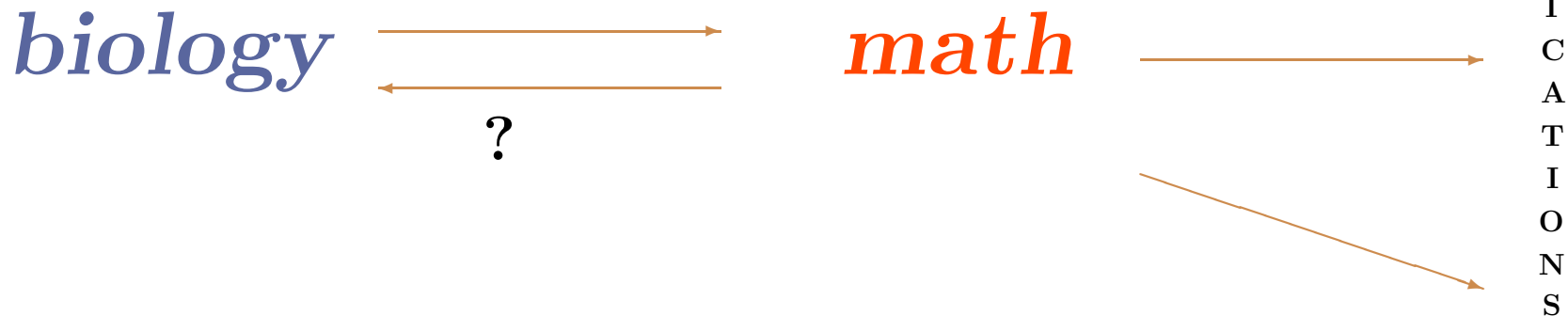
Sponsored by the Joint Policy Board for Mathematics:
American Mathematical Society • Mathematical Association of America • Society for Industrial and Applied Mathematics

Math & Bio

physics classical source of math concepts



very probably in the future:



of course, the hope is for “←”

but in any case, great *inspiration for new mathematics!*

The Role of Math Modeling

need for mathematical models long recognized

- 1950s Cybernetics, Wiener, Ashby; “homeostasis”
- dynamics, feedback, of molecular networks in 1970s (e.g. Tyson, Othmer, Kauffman, Savageau, Glass)

what is different now:

- lots of data
- new experimental design techniques
 - gene knock-outs, mutations, mass copying (PCR)
 - measurement tools (“outputs”): gene chips, GFP

allow hypotheses testing & falsification

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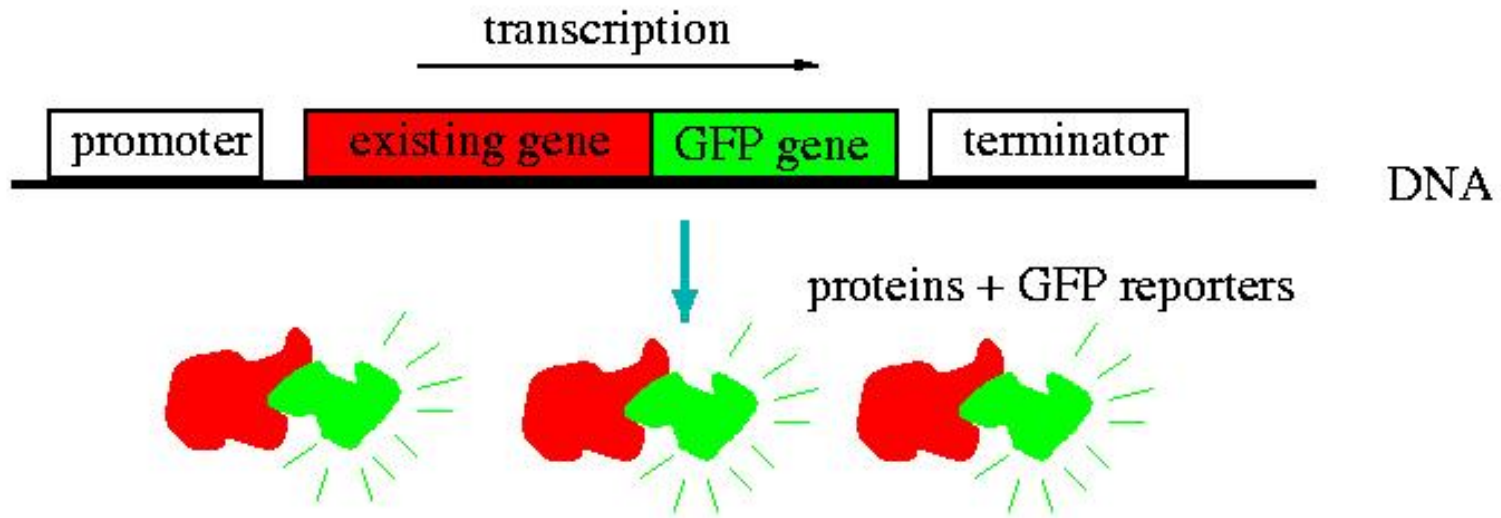
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Scalar Outputs: GFP

Green Fluorescent Protein gene inserted adjacent to gene coding for protein to be measured



genes are expressed simultaneously,
measure intensity of the GFP light emitted

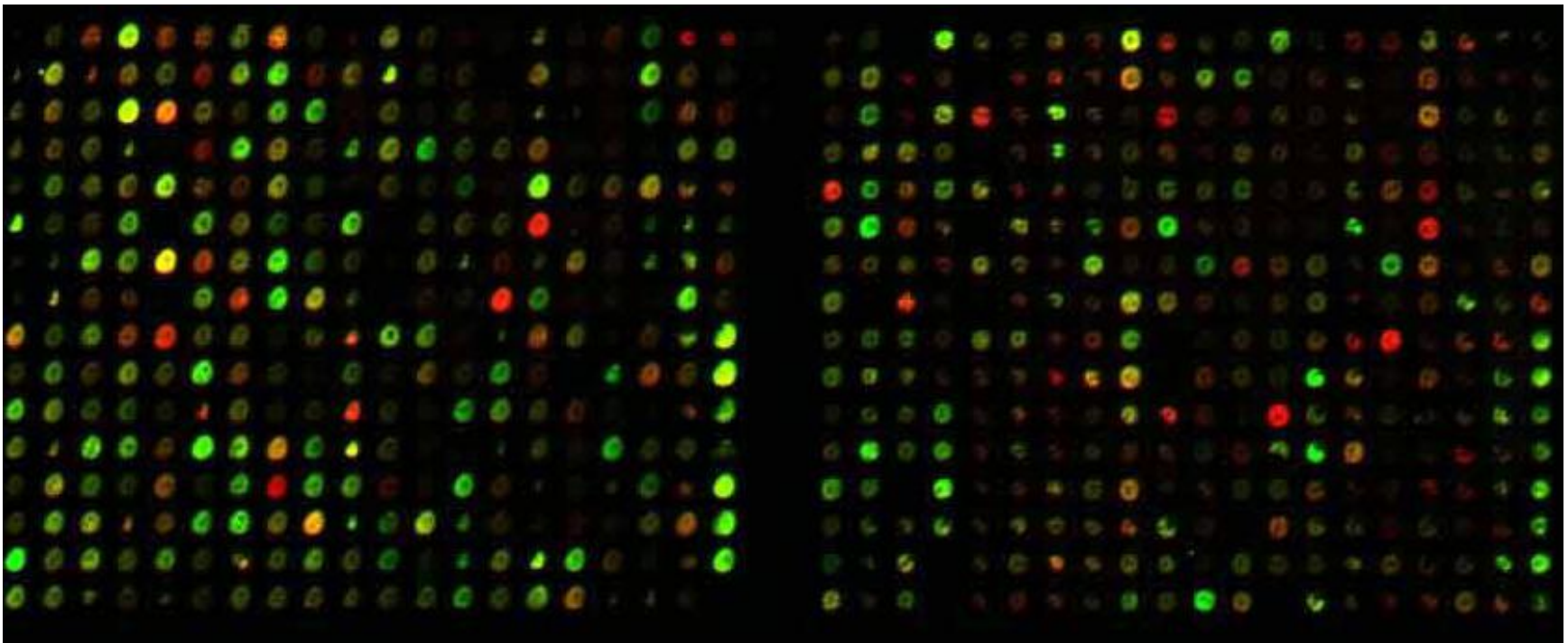
source of GFP:
Aequoria victoria,
Pacific NW jellyfish



GFP = aequorin
used as “reporter”
green fluorescence
when UV irradiated

Vector Outputs: Gene Chips

gene arrays $\rightsquigarrow$ activity levels of many genes
spread sample on an array of “tuned” detectors
 $\rightsquigarrow$ more complete snapshot of the current state of the system



activity levels of thousands of genes; technology: estimate mRNA (proportional to protein);

too unstable, so reverse-transcribed to complementary DNA; spread sample on chip; binds specific DNA at each spot

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Challenging Questions

- *information-processing (i/o)?*
- *signal transduction pathways?*
- *reverse engineering (inverse problem)*
 - parameters (reaction constants)?
 - protein expression levels (state $\in \mathbb{R}^{100,000}$)?
- *what “modules” appear repeatedly?*
- *why cascades and feedback loops?*
- *dynamical properties?*
 - stability, oscillations, ...
- *how to **control** using external inputs?*

Focus Today: Adaptation

regulation and “adaptation” at intracellular biochemical level are central to cellular life

new mathematical questions arise, compared to most common analogues in engineering

a few “short stories”:

- “negative feedback” (inhibition): special meaning
- internal model principle with signal detection
- global attractors, lack of oscillations deduced from network topology (parameter-indep stability)
- cascades and feedback, small gain

many $\supseteq$ sophisticated experimental work

Other Work, not Covered

- *global observers for concentrations*
in chemical networks motivated by biology (w/Chaves, EJC)
- *determination of network structure*
signaling network topology (w/ Kholodenko et. al., PNAS)
- *parameter identification*
under constrained input classes such as steps (J. Nonlinear Sci.)

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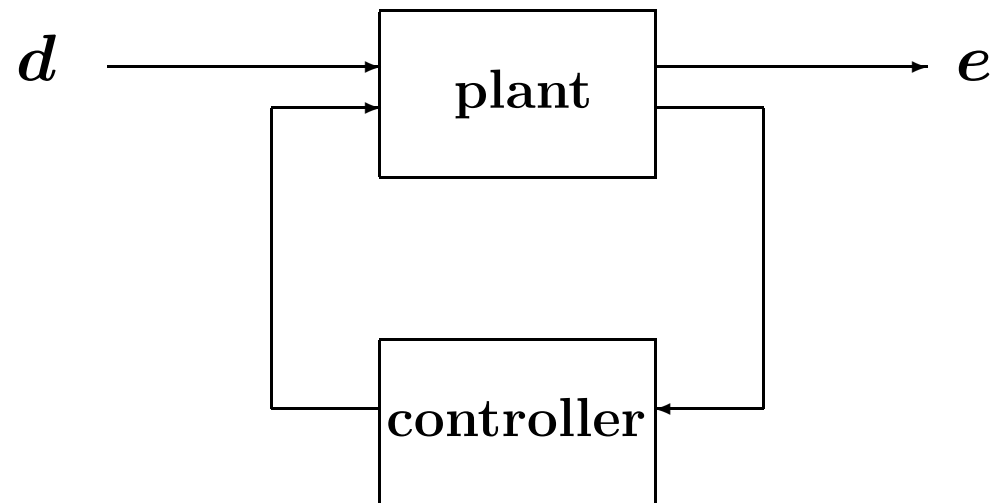
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Disturbance Attenuation

in engineering, negative feedback typically used
in order to decrease uncertainty due to perturbations

↘ ISS (or H_∞ , etc) gain of “disturbance $\mapsto$ error” operator



basic principle: higher gain $\Rightarrow$ smaller error

next: a nontrivial experimental verification of this principle,
in a cell, through genetic engineering

in *E.coli*, about 40% of transcription factors self-regulate

TetR, or tetracycline repressor protein, defends *E.coli* against tetracycline, and is a major source of antibiotic resistance

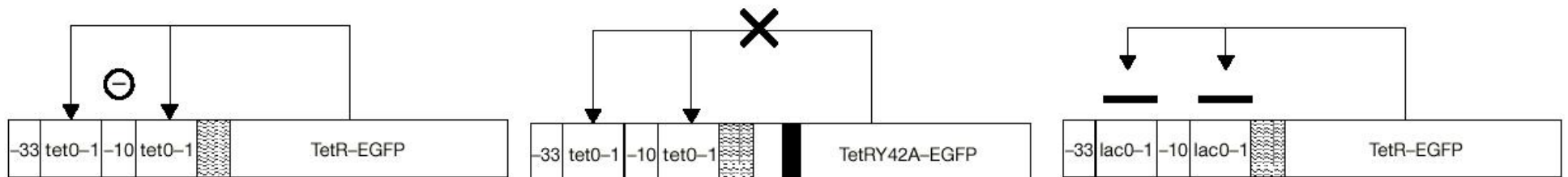
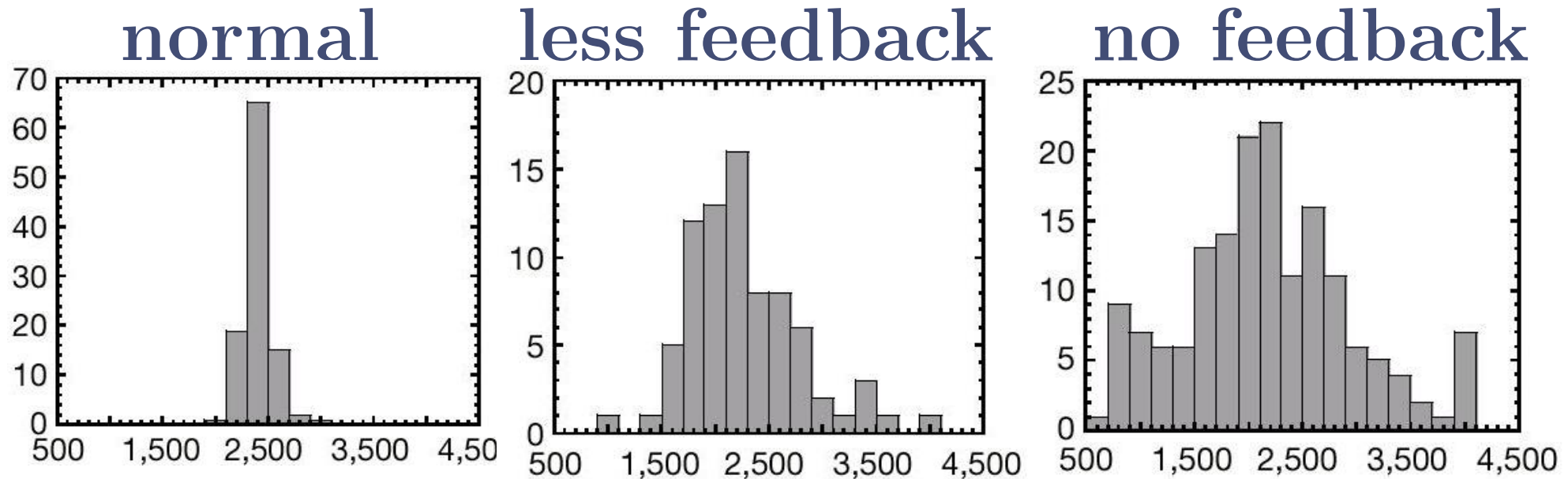
TetR regulates its own formation through a feedback loop



protein level measured by splicing GFP gene to gene coding for TetR
experiment: “loop gain” $\searrow$ by mutations/replace operator
(feedback loop partially or almost totally disabled)

lower “gain” $\Rightarrow$ more variability in protein expression

Experimental results



“negative feedback” in biology typically *not* additive “ $x - ke$ ” (!)
 but rather inhibition: multiplication by small factor $\left(\frac{1}{1+ke}\right) x$

coefficient of variation (var/μ)

internal noise (small copy) vs external noise

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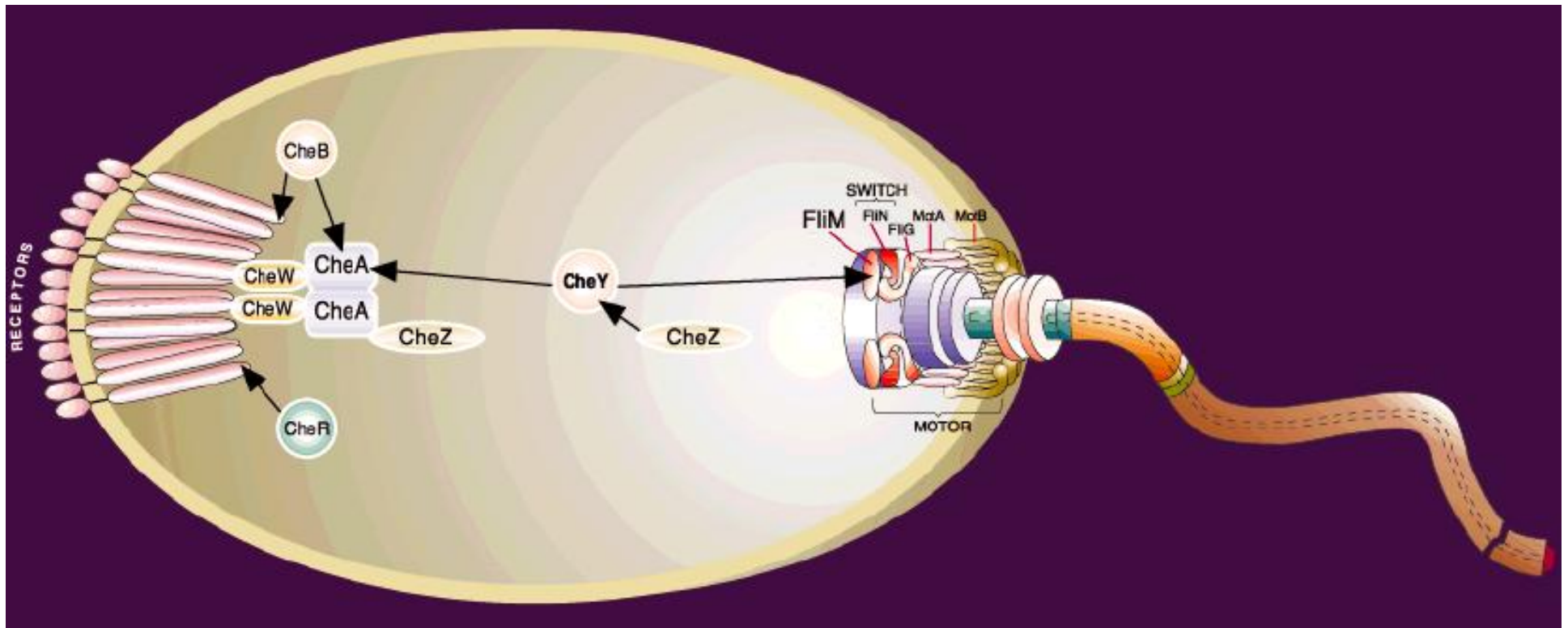
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E.coli Chemotaxis

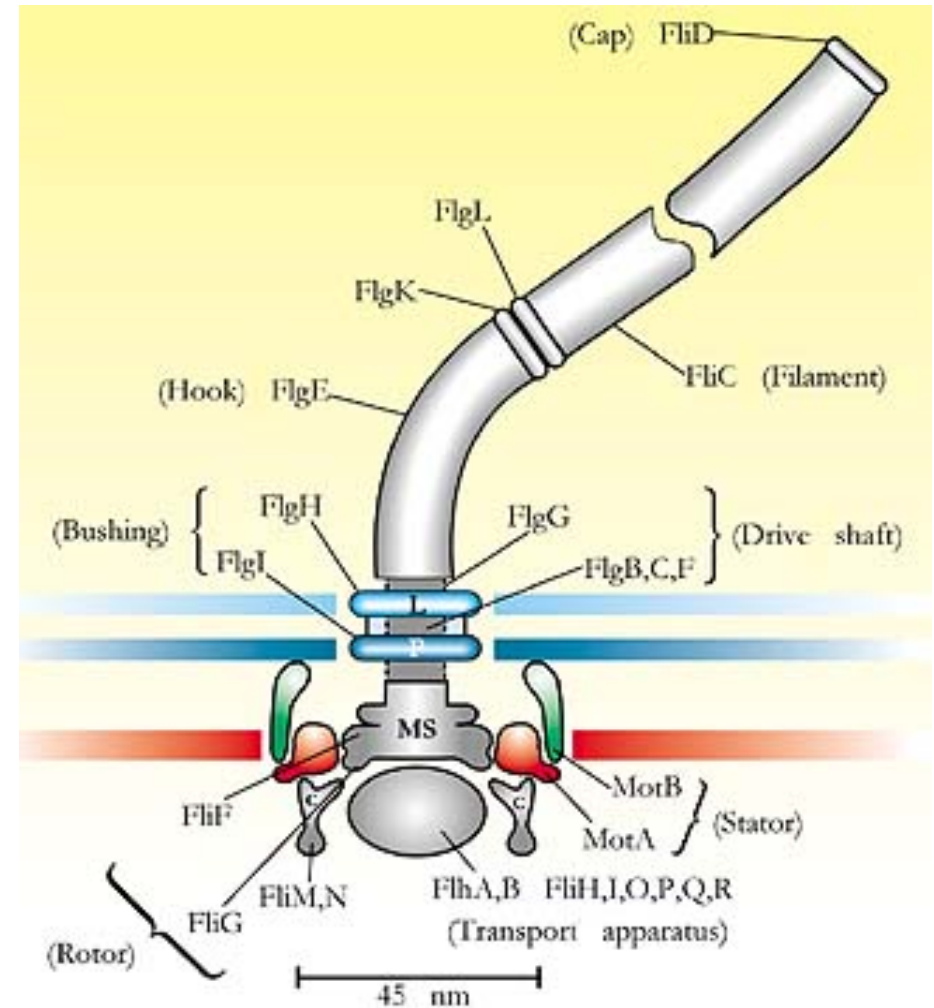
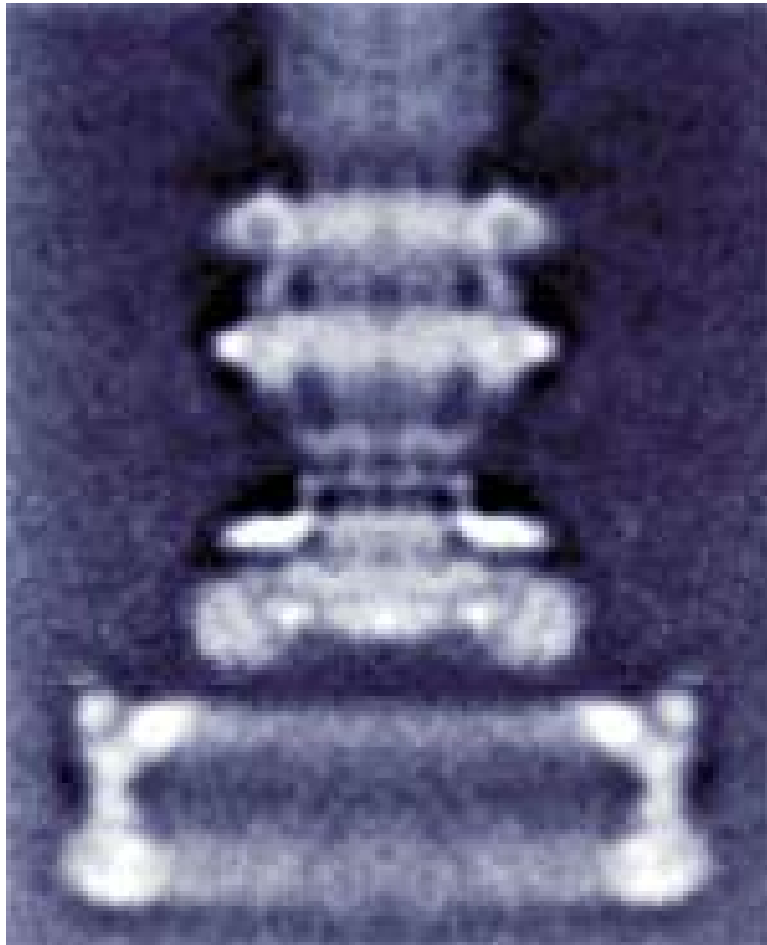
E.coli moves (**taxis**), propelled by flagella, in response to **chemical** attractants/repellents



nutrient level → signaling system → signal to motors

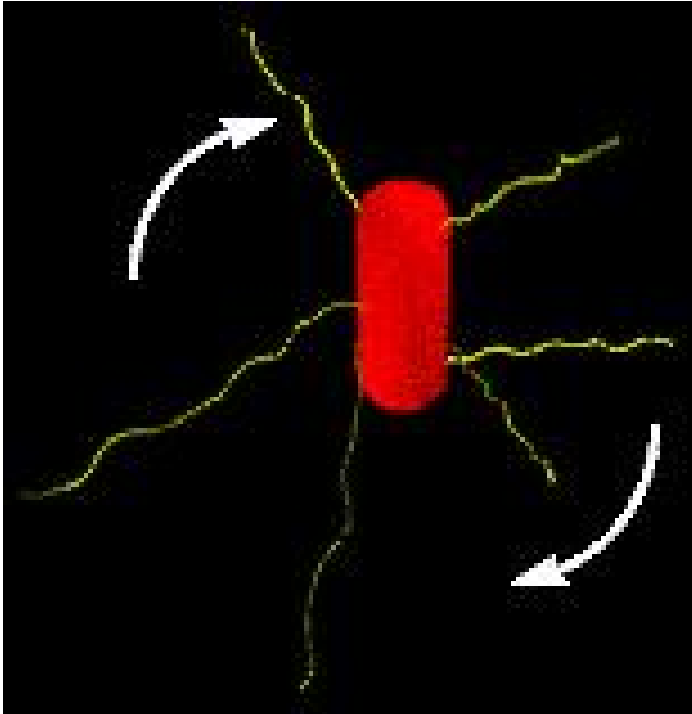
Berg: typical cell has up to six flagella and hundreds of porins (channels for nutrients); cell body is 2 μm long

Motor: Micrograph, Diagram



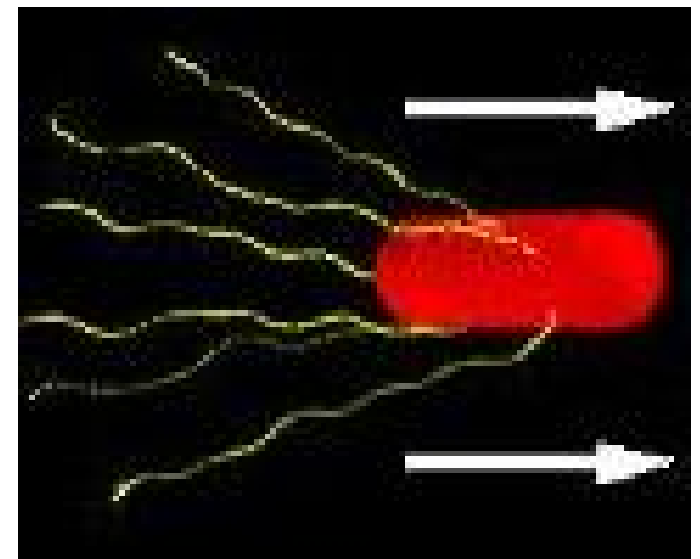
engines, propellers, particle counters, rate meters, gear boxes
“a nanotechnologist’s dream” (Berg, Physics Today, 2000)

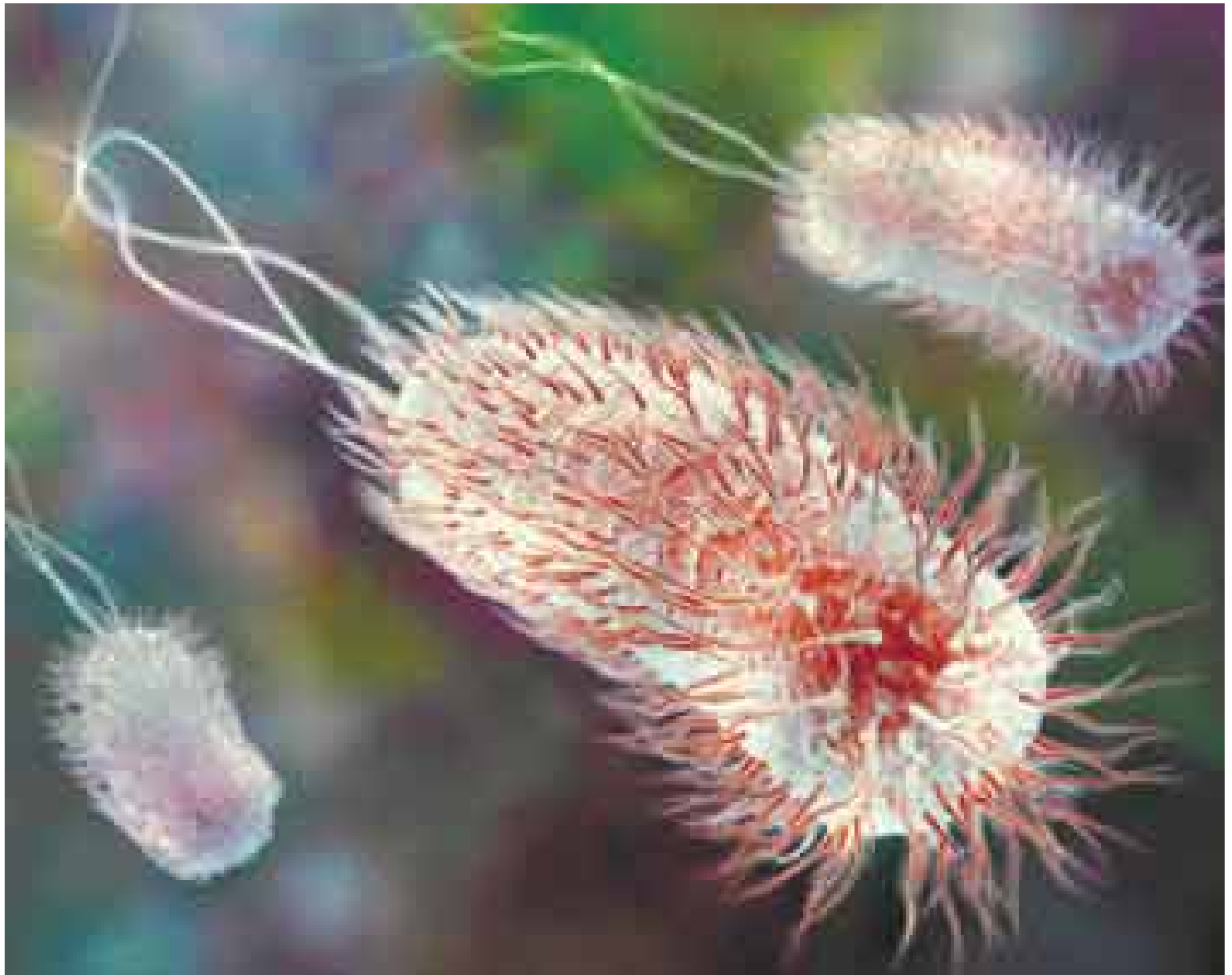
Rotation Sense & Movement



Mode 1:
tumbles if flagella
turn clockwise

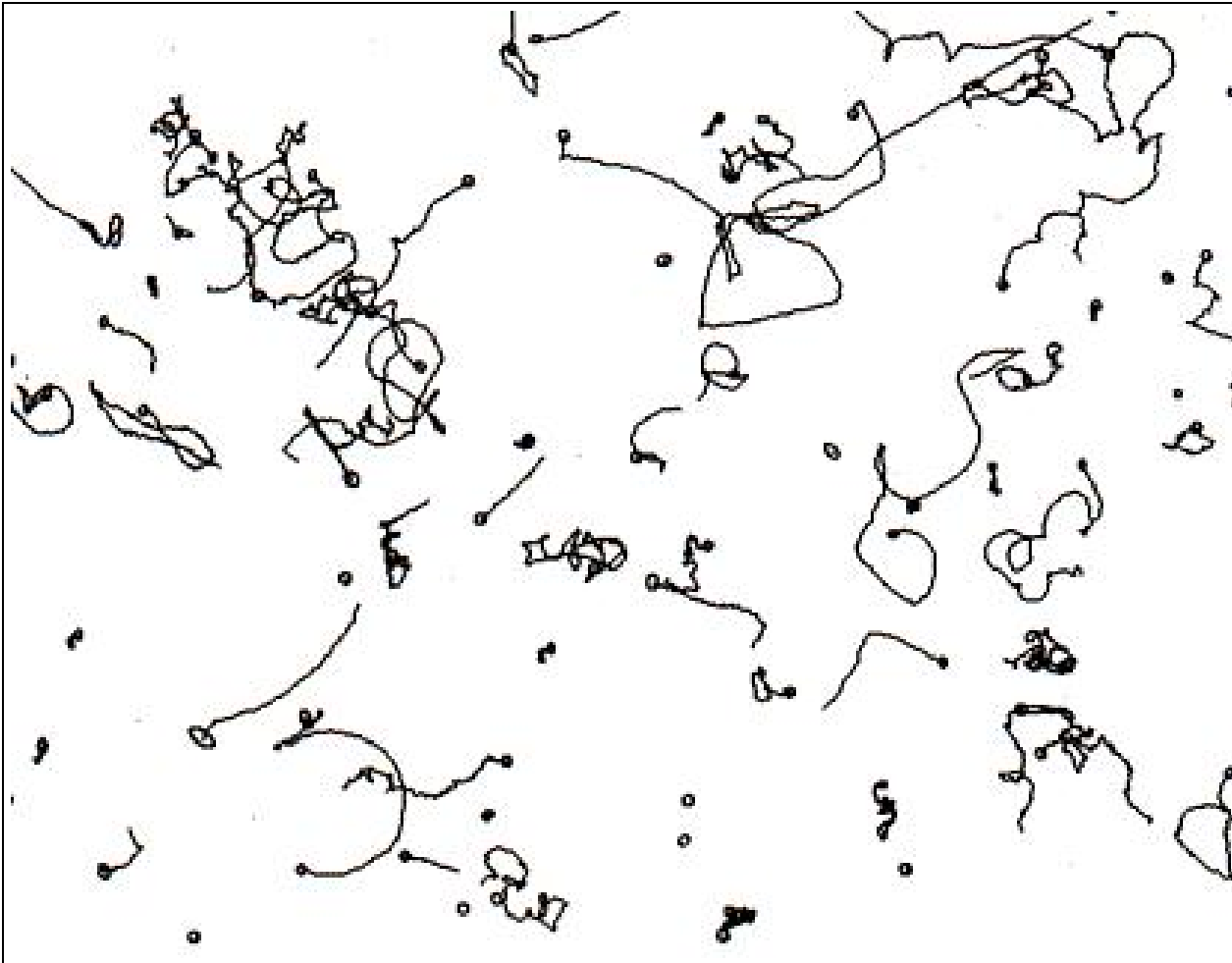
Mode 2:
runs if flagella turn CCW
(they form a bundle)





Stochastic Gradient Search

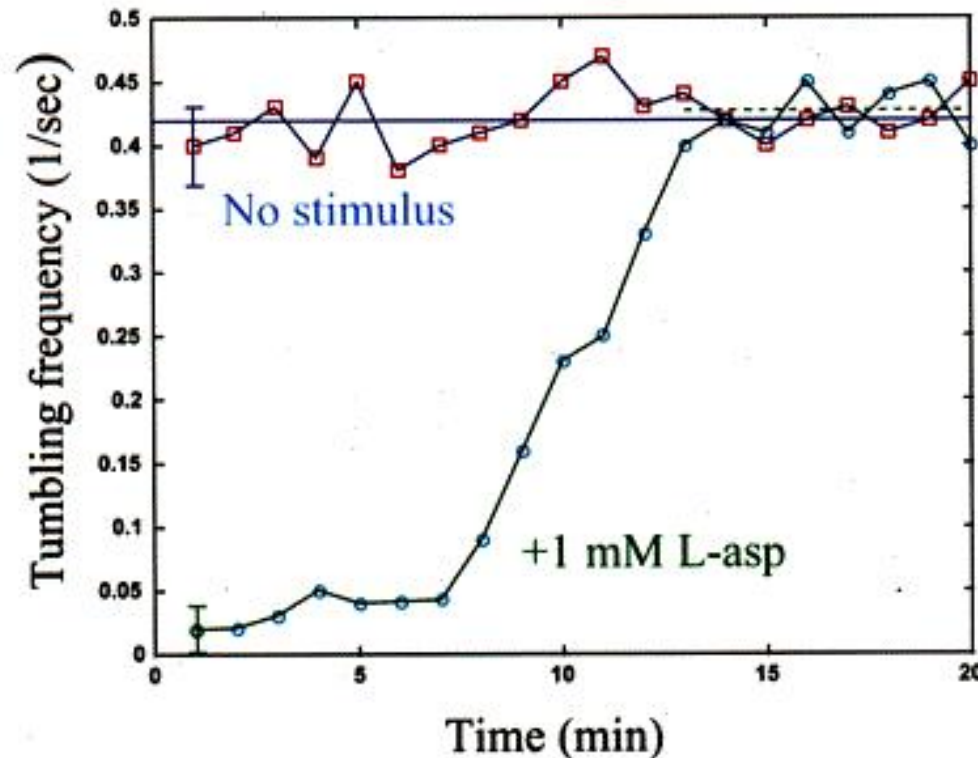
sense increased concentration $\rightarrow$ stop tumbling (and run)
 ∇ concentration $\approx 0 \rightarrow$ resume tumble (“search mode”)



runs caused by filaments coordinated in bundle (turning CCW); tumbles occur when they turn CW, resulting in independent moves (lack of symmetry). runs are biased, drifting about 30 deg/s due to viscous drag and asymmetry; no inertia (low Reynolds); mean run interval about 1 s, mean tumble interval about 0.1 s

nutrient level $\longrightarrow$ signaling system $\longrightarrow$ signal to motors

- transient signal (“stop tumbling, run toward food”) issued in response to a change in concentration
- adapt, signal converges to default (“OK, tumble”)



input $\Rightarrow$ transient “stop tumbling signal” ($y(0) \neq y^*$), but $y(t) \xrightarrow[t \rightarrow \infty]{} y^*$
(experiments by Alon, Surette, Barkai, Leibler *Nature* 1999)

Not High-Gain Feedback

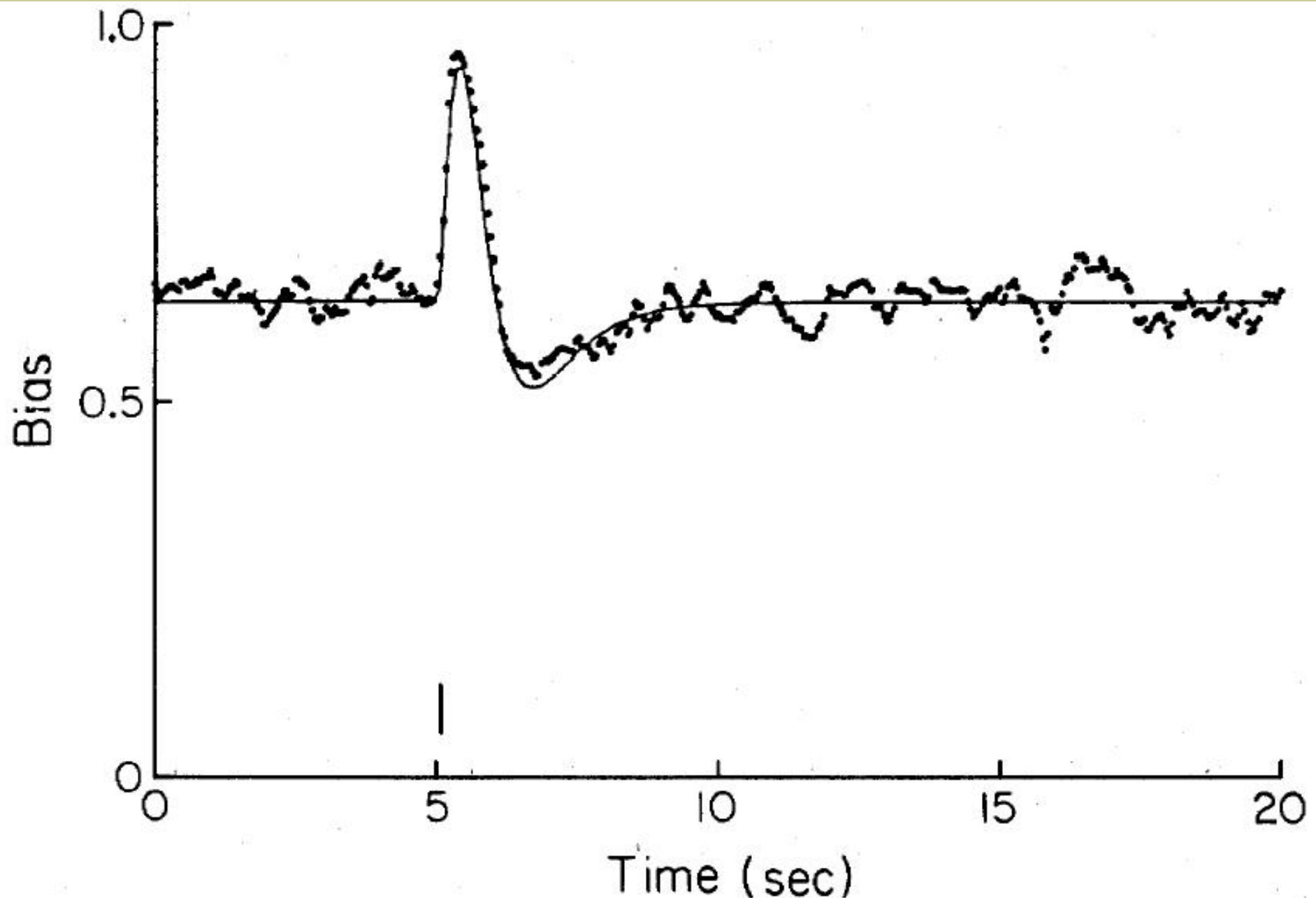
high sensitivity in response to changes
indicates that not all signals attenuated

rather, behavior consistent with regulation with respect to
constant signals (chemical gradient) plus nontrivial I/O

indeed, experimental measurements of impulse-response
(local, small amplitude, wrt default tumbling signal)
fit perfectly to transfer function having a zero at $s = 0$

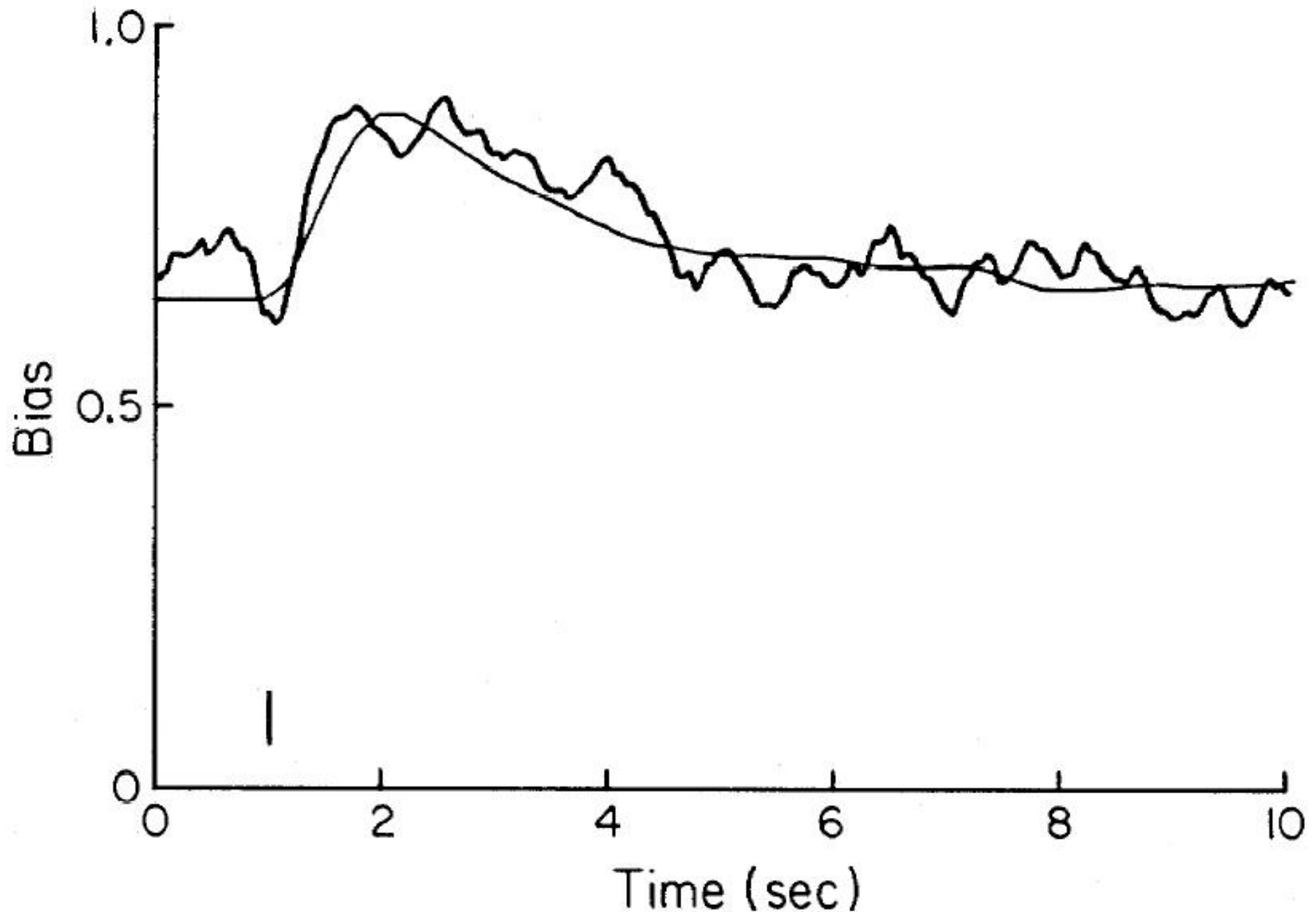
further, integral of i/r fit is itself fit perfectly by data:

Zero DC Gain to $y - y^*$



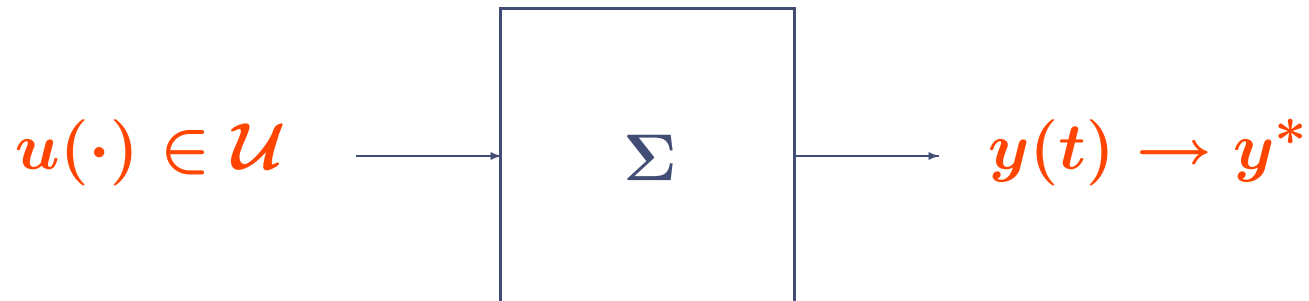
impulse response (population avg), Segall, Block, Berg, PNAS 1986

Step Response, Indeed $\int$

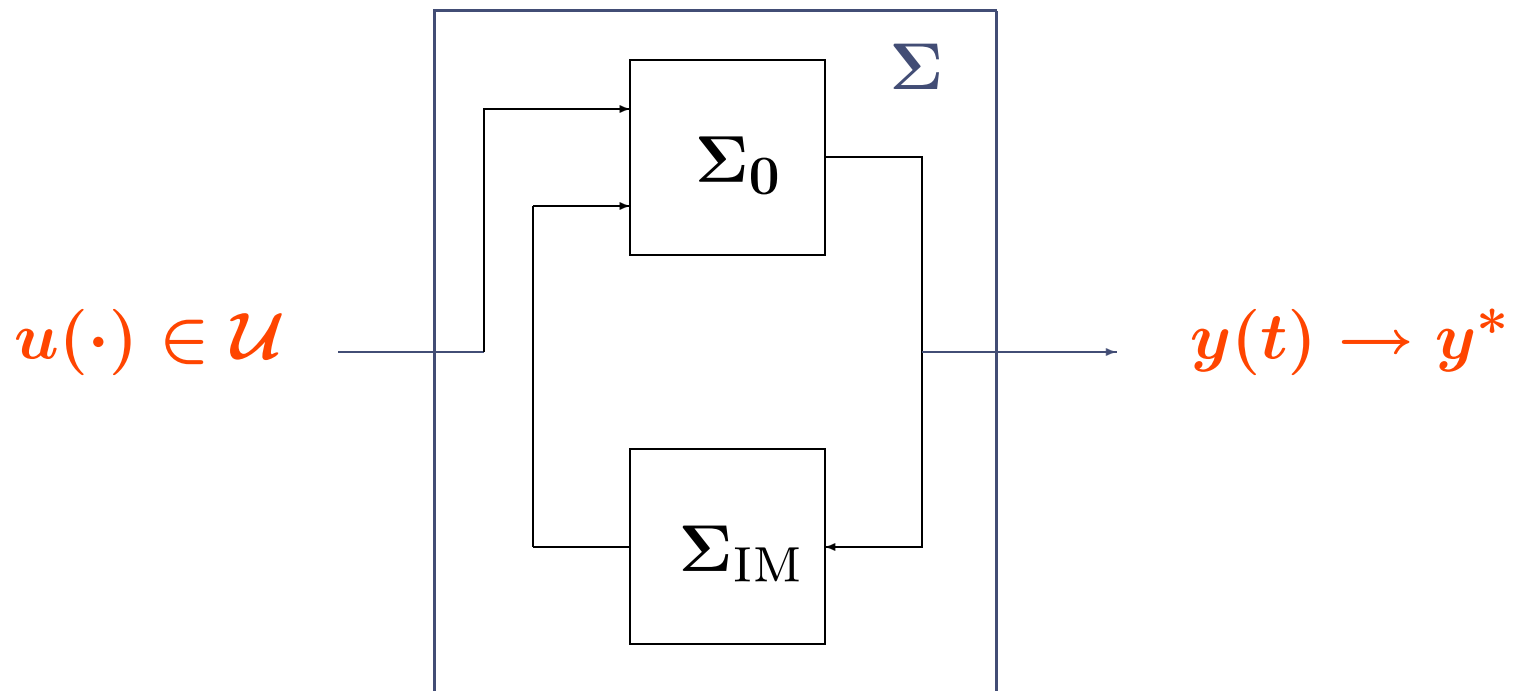


step response (population avg), Segall, Block, Berg, PNAS 1986

Internal Model Principle



$\Rightarrow$ decomposition:



where the subsystem Σ_{IM} is driven by the “error” $y(\cdot) - y^*$ and can generate all possible “disturbances” $u(\cdot) \in \mathcal{U}$

Bio IMP Technically $\neq$ Usual

i.e.: if system Σ regulates w.r.t. all inputs $u(\cdot) \in \mathcal{U}$
($\mathcal{U}$ = some predetermined class, e.g. constants)

then Σ *must* contain an IM generating $\mathcal{U}$
(intuitively, guess u based on $y - y^*$, cancel effect)

assumptions needed (else: $\Sigma \equiv y^*$ adapts w.r.t. to any $\mathcal{U}$)
typical theorem: “robustness” (Wonham, Francis, Hepburn)
“robust” = structurally stable/Whitney topo, model = flow embedding
Wonham-Francis 1970s: linear; Wonham-Hepburn 1981: Center Mfld’s

note: small transient (overshoot) desirable in engineering
— e.g. cruise-control, active suspension

but chemotaxis $\neq$!! *want* to detect sudden change in $u(\cdot)$

$\therefore$ desirable theorem:

signal detection and adaptation $\Rightarrow$ internal model

linear: pole/zero cancel s.t. cascade stable $\Rightarrow$ poles of exosystem generating u ’s $\subseteq$ zeroes of $\Sigma \subseteq$ poles of feedback subsystem

One Theorem

(EDS, 2002)

suppose $\dot{x}(t) = f(x(t)) + u(t)g(x(t))$, $y(t) = h(x(t))$
adapts to $\mathcal{U}$ = generated by Poisson-stable exosystem

f & g smooth vf's; h smooth $\mathbb{R}^n \rightarrow \mathbb{R}$, $f(0) = h(0) = 0$

assume finite uniform relative degree:

$$(\exists r) \quad L_g L_f^k h \equiv 0 \quad \forall k < r-1, \quad L_g L_f^{r-1} h(x) \neq 0 \quad \forall x \in \mathbb{R}^n$$

and vf's $\tau_i := \text{ad}_{\tilde{f}}^{i-1} \tilde{g}$, complete and pairwise commutative

$$\tilde{g}(x) = \frac{1}{L_g L_f^{r-1} h(x)} g(x), \quad \tilde{f}(x) = f(x) - \left(L_f^r h(x) \right) \tilde{g}(x),$$

(linear systems: $\equiv$ nonzero transfer function)

then $\exists$ diffeomorphism $\rightsquigarrow$

$$\begin{aligned} \dot{z}_1 &= f_1(z_1, z_2) + u g_1(z_1, z_2), & y &= \kappa(z_1) \\ \dot{z}_2 &= f_2(y, z_2) \end{aligned}$$

and $\exists$ smooth scalar $\varphi(z_2)$ s.t., $\forall u \in \mathcal{U}$, $\exists$ solution of

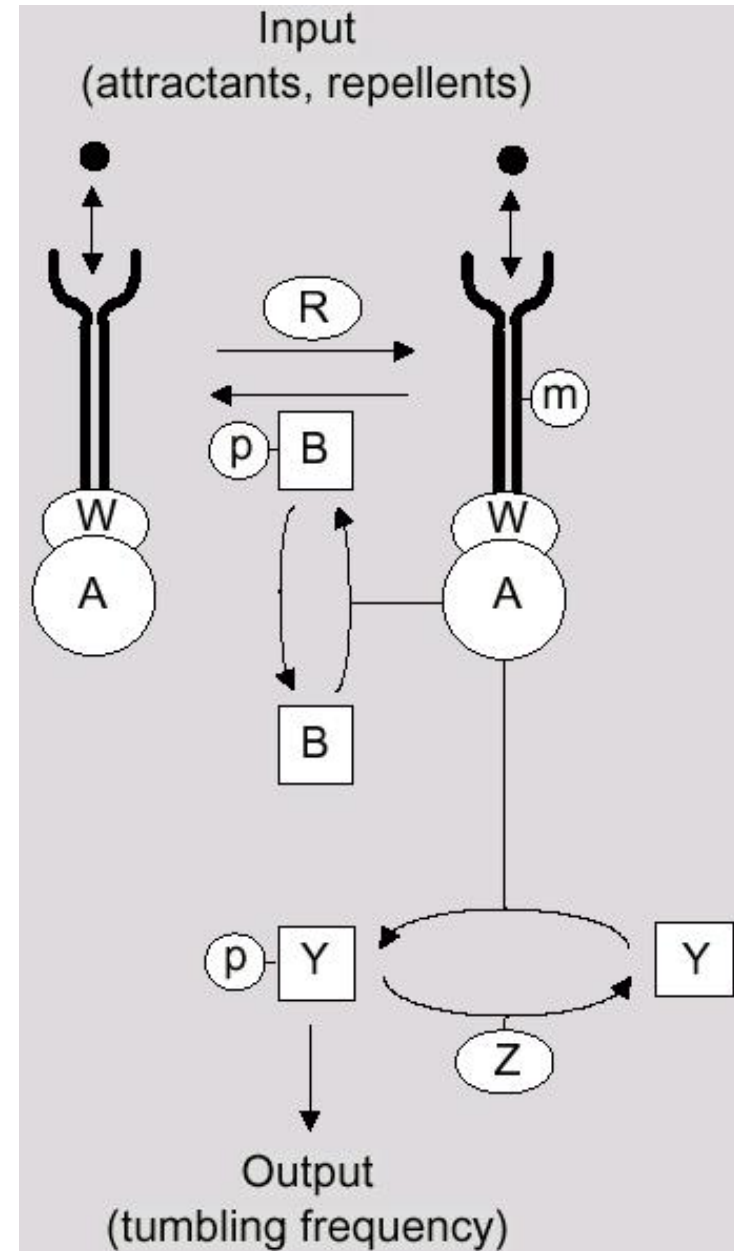
$$\dot{z}_2 = f_2(0, z_2) \quad \text{s.t.} \quad \varphi(z_2(t)) \equiv u(t)$$

Internal Model (Integrator)

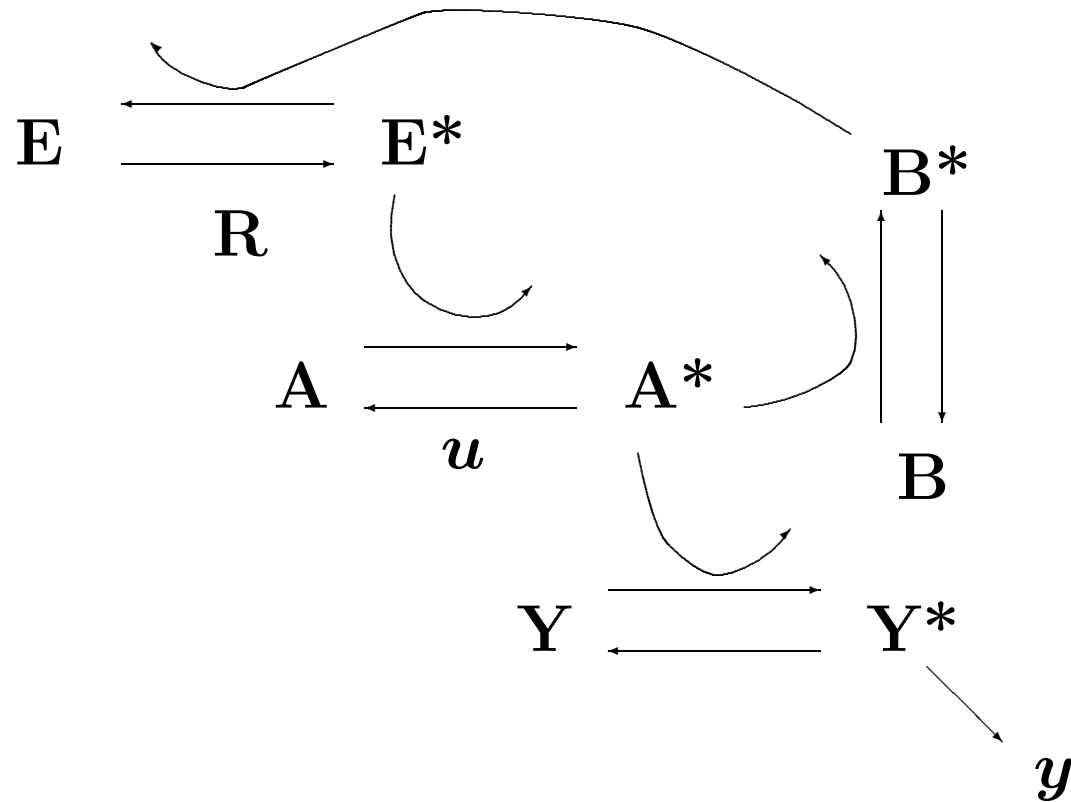
Yi/Huang/Simon/Doyle
(PNAS 2000):

IMP $\Rightarrow$ integral feedback
in molecular mechanism of
Barkai&Leibler, Nature 1997

(also: Iglesias & Levchenko,
CDC 2001)



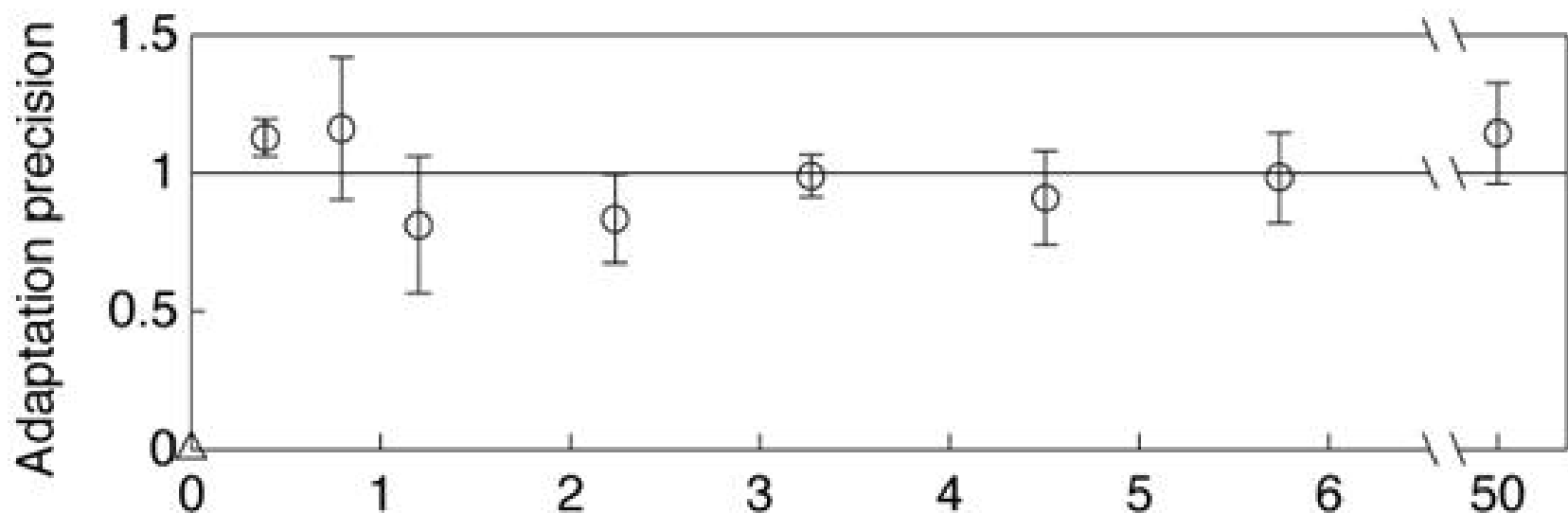
One Interpretation



A, B, Y, E are che-A, che-B, che-Y, and receptors; stars = active forms (phosphorylation, methylation level); assume saturation of R, E^* , so $(e=E) \dot{e} = \alpha b - \beta r$ and hence in steady state $b(\infty) = \beta r / \alpha$ (A^* and so Y^* proportional)
 did not show che-W (at receptor) nor che-Z ($Y^* \rightarrow Y$); R is constant

Robustness of Adaptation

integral feedback confers robustness of adaptation behavior,
with respect to appropriate system parameters (“plant”)
experiments (Alon et al, 1999): CheR 100-fold, $=\lim e(t)/y^*$



global robustness of stability behavior may be
desirable in biology, where parameters
could easily range over orders of magnitude

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- measurement noise & hybrid systems

■ *discussion*

An Excursion into Immunology

immune system charged with destruction and elimination of invading organisms & toxic products they produce



T-cell attacking influenza virus-infected target

“T” (thymus) lymphocytes (type of white blood cells) kill virus-infected [or mutated, etc] cells including macrophages, which ingest antigens

Kinetic Proofreading, McKeithan *PNAS*'95

in infected cells (APC = “antigen presenting cell”)
MHC molecules present foreign antigens on cell surface
(peptide fragments of foreign proteins)

must distinguish *foreign* vs *self* (autoimmune diseases!)

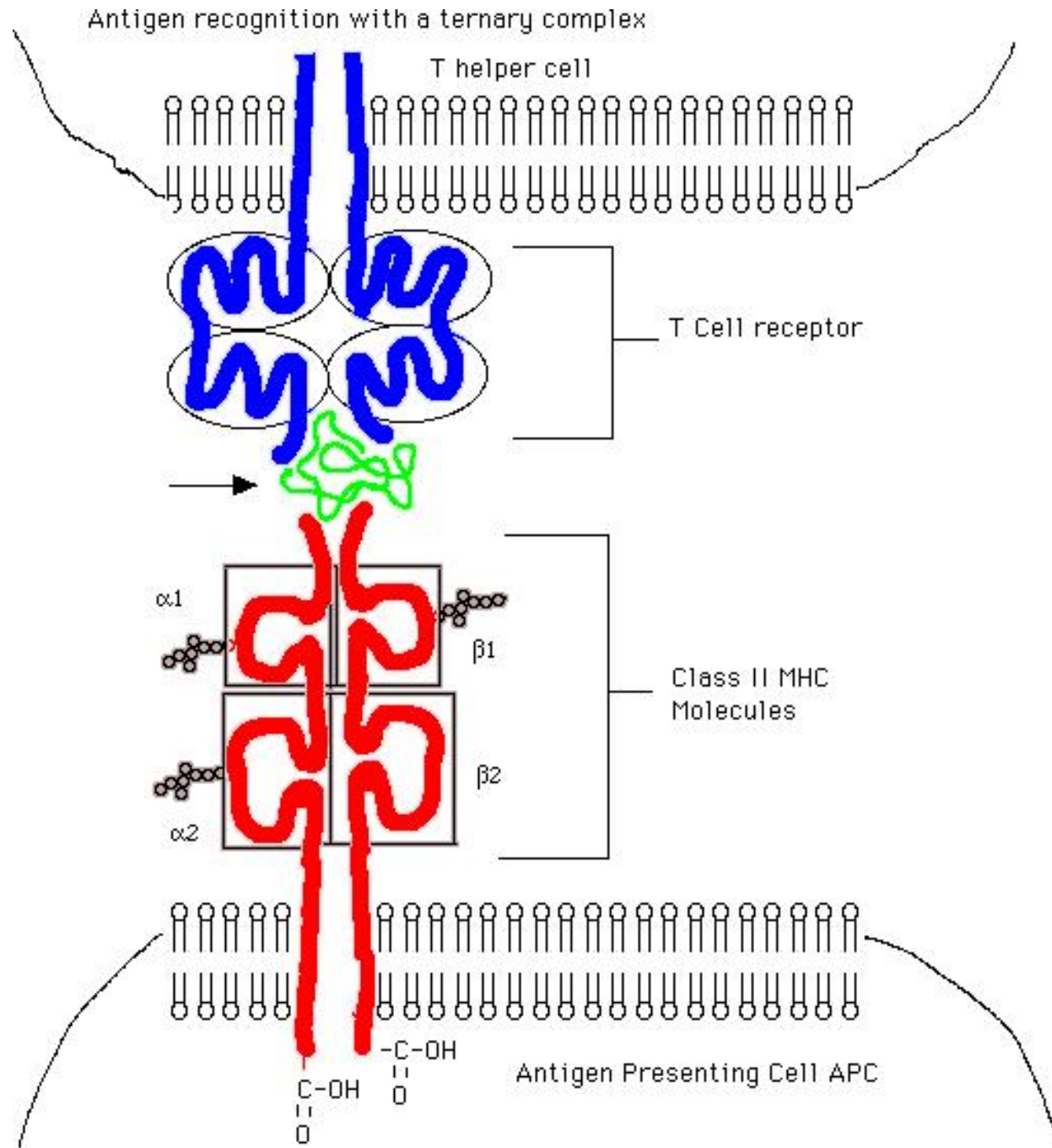
immune system distinguishes *extremely similar* antigens
(proteins differing in single aminoacid, optical isomers)

one suggestion for how selectivity is accomplished:

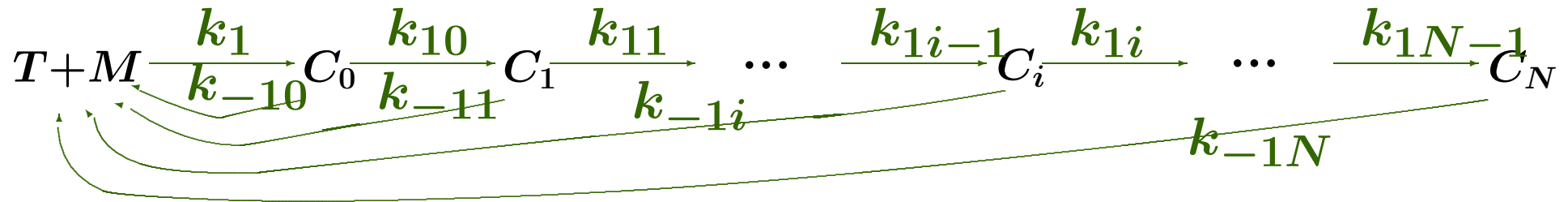
kinetic proofreading: cascade lower-selectivity steps,
with reverse error correction

“chain of modifications of T-cell receptor complex
via tyrosine phosphorylation and other reactions
↪ increased response sensitivity & selectivity”

APC & T-Cell



McKeithan's Reaction



$T(t)$ = concentration of T-cell receptor (TCR)

$M(t)$ = peptide-major histocompatibility complex (MHC)

$C_0(t)$ = initial ligand-receptor complex

$C_1(t), \dots, C_N(t)$'s = intermediate complexes

k_1 = association rate for reaction producing initial ligand-receptor complex C_0 from TCR/MHC

$k_{1,i}$ = other reaction (phosphorylation,...) rates

$k_{-1,i}$ = dissociation rates

McKeithan postulated recognition signal $\sim C_N(+\infty)$

analyzed uniqueness of steady states,

under simplifying assumptions $k_{1,i} \equiv k_1$ & $k_{-1,i} \equiv k_{-1}$

Motivates Theory Questions

- *structure of steady states, in general?*
unstable $\Rightarrow$ little biological significance, so:
- *stability? (periodic orbits? chaos? ..?)*
- *behavior depends on parameter values?*
- *fits in general theoretical framework?*

$\leadsto$ general framework for stability of (some, not all!)

chemical-like systems subject to conservation laws; e.g.

phosphorylated + unphosph forms of protein $\equiv$ constant

(compare: Hamiltonian formalism in mechanics)

General: Chemical Networks

$$\dot{x} = f(x) = \sum_{i=1}^m \sum_{j=1}^m a_{ij} x_1^{b_{1j}} x_2^{b_{2j}} \dots x_n^{b_{nj}} (b_i - b_j)$$

constants a_{ij} all ≥ 0 , matrix $A = [a_{ij}]$ [block] irreducible

$m \leq n$ ($= \#$ species = dimension of system)

each $b_j \in \mathbb{R}^n$ w/entries $b_{1j}, b_{2j}, \dots, b_{nj}$, integers ≥ 0

$B := [b_1, b_2, \dots, b_m]$ has rank m , no row vanishes

seen as system on $\mathbb{R}_+^n$ (forward invariant)

“zero deficiency and weakly reversible” networks
studied by Feinberg, Horn, and Jackson in 1970s

Some Results

(EDS, IEEE TAC'01)

for all possible values of kinetic constants:

- *equilib set E embedded submanifold*
- *foliation into invariant sets $\pitchfork E$*
- $\forall x(0), x(t) \rightarrow E \text{ as } t \rightarrow +\infty$
- *GAS rel to leaves, for positive states*

rules out periodic trajectories, chaos

(contrast Belousov-Zhabotinsky, Prigogine, etc)

also: estimate robustness to unmodeled dynamics

and extensions to (some) non-ideal mass action kinetics

and various control problems: stabilization, observers, ...

relies upon deep results & tools due to Feinberg et. al.

Very Simple Example

e.g.: kinetic proofreading
with $N = 0$: $T + M \rightleftharpoons C_0$
 $x_i(t)$ = concentrations of
 T, M, C_0 , eqs:

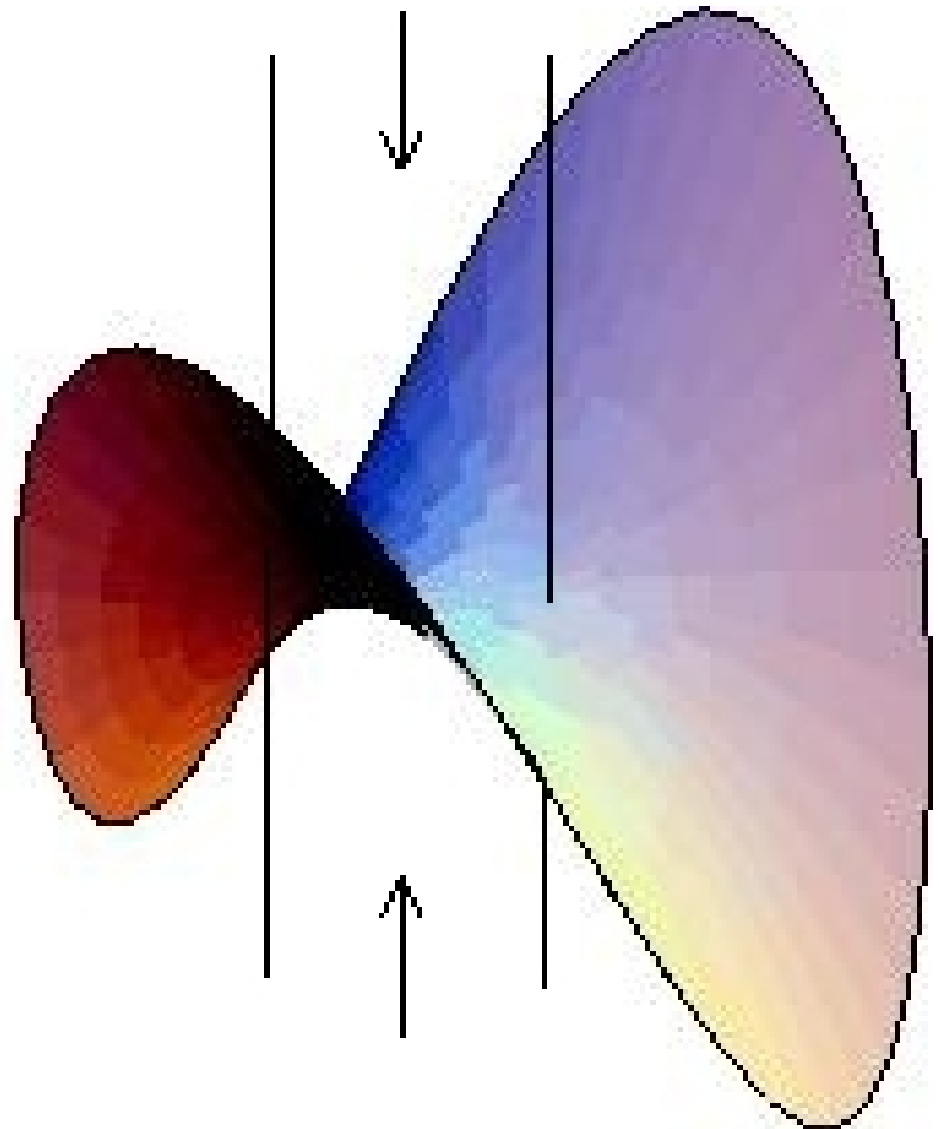
$$\dot{x}_1 = -k_+ x_1 x_2 + k_- x_3$$

$$\dot{x}_2 = -k_+ x_1 x_2 + k_- x_3$$

$$\dot{x}_3 = k_+ x_1 x_2 - k_- x_3,$$

steady states $\sim$
hyperbolic paraboloid
 $k_- x_3 - k_+ x_1 x_2 = 0$

global convergence, along each line, to equilibria



Outline

■ *introduction: biochemistry*

- central dogma
- life $\sim$ protein [and gene] networks
- mathematical modeling - why now?
- measurement technologies
- (I/O)systems-theoretic questions

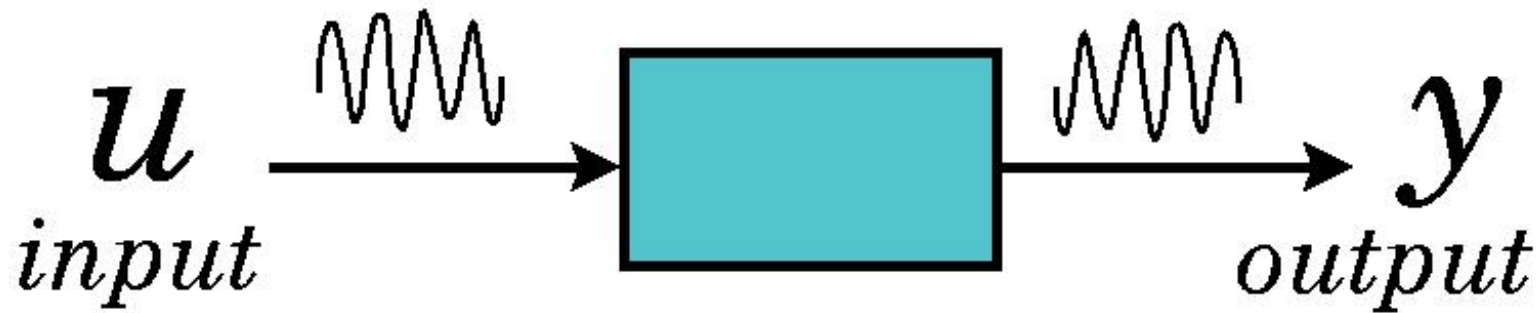
■ *focus: “adaptation” & “robustness”*

- inhibitory feedback in gene regulation
- integral control in chemotaxis
- robust stability structures; kinetic proofreading
- **interconnections, small gain, & MAPK cascades**
- measurement noise & hybrid systems

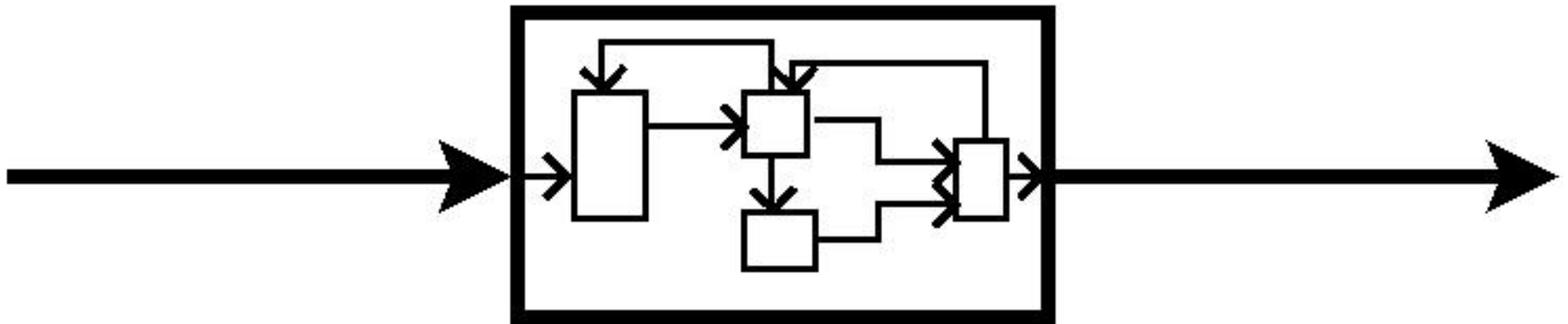
■ *discussion*

Interconnections

in broad terms, control systems theory deals with interconnections of devices which process input signals into output signals



basic principle: decompose (finer resolution) into “simpler” subsystems & study emergent properties of interconnections



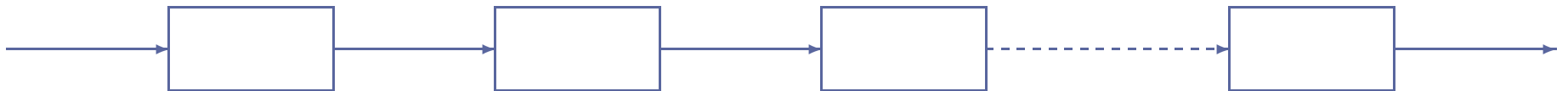
Cascades

the analysis of properties of interconnections of I/O systems is the “bread and butter” of systems theory

interest in signaling properties, stability, etc

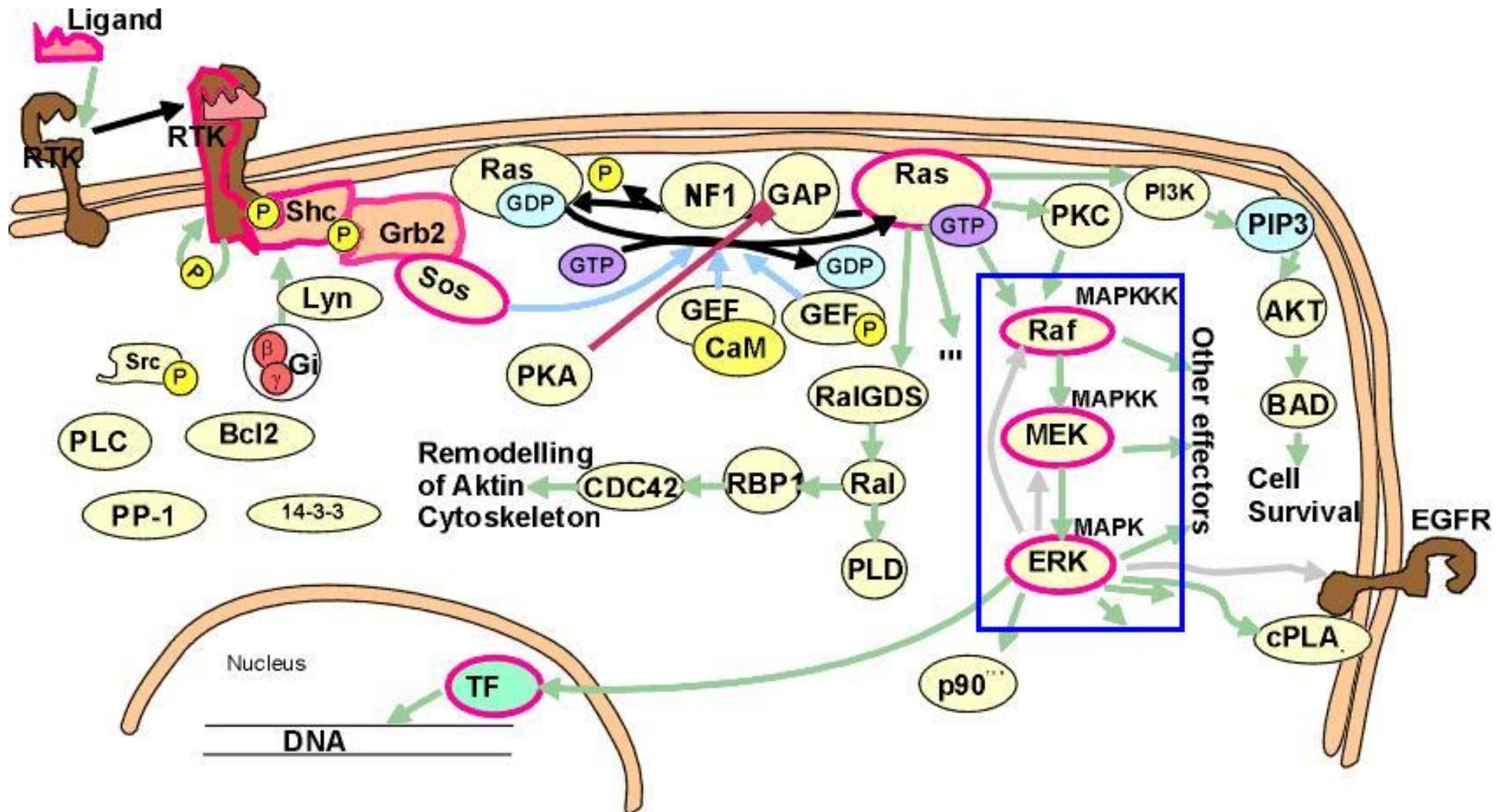
■ *first step: cascades*

(preservation of stability $\leftrightarrow$ ISS 😊)



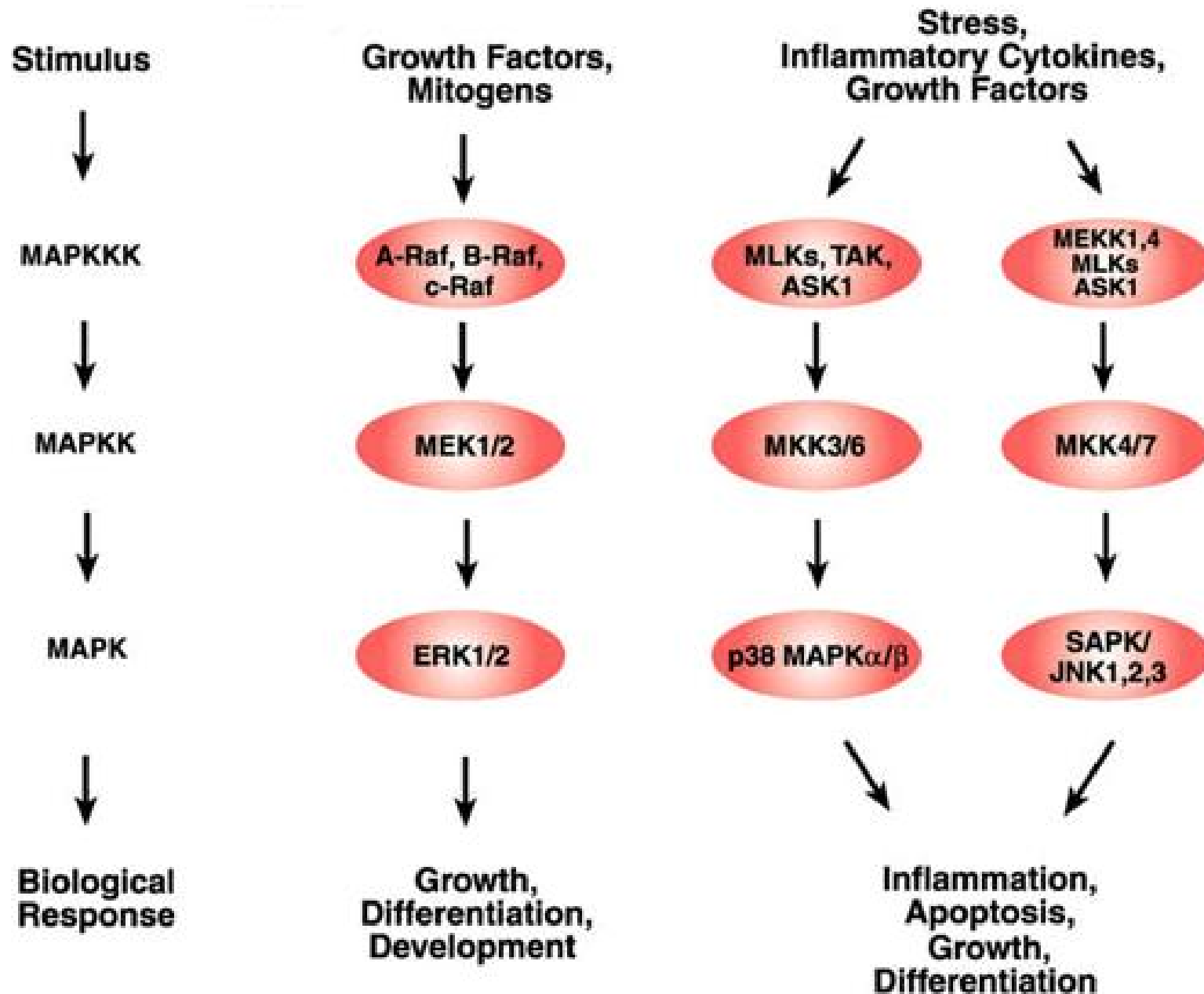
■ *then: feedback*

MAPK Cascades (e.g.: RAS)

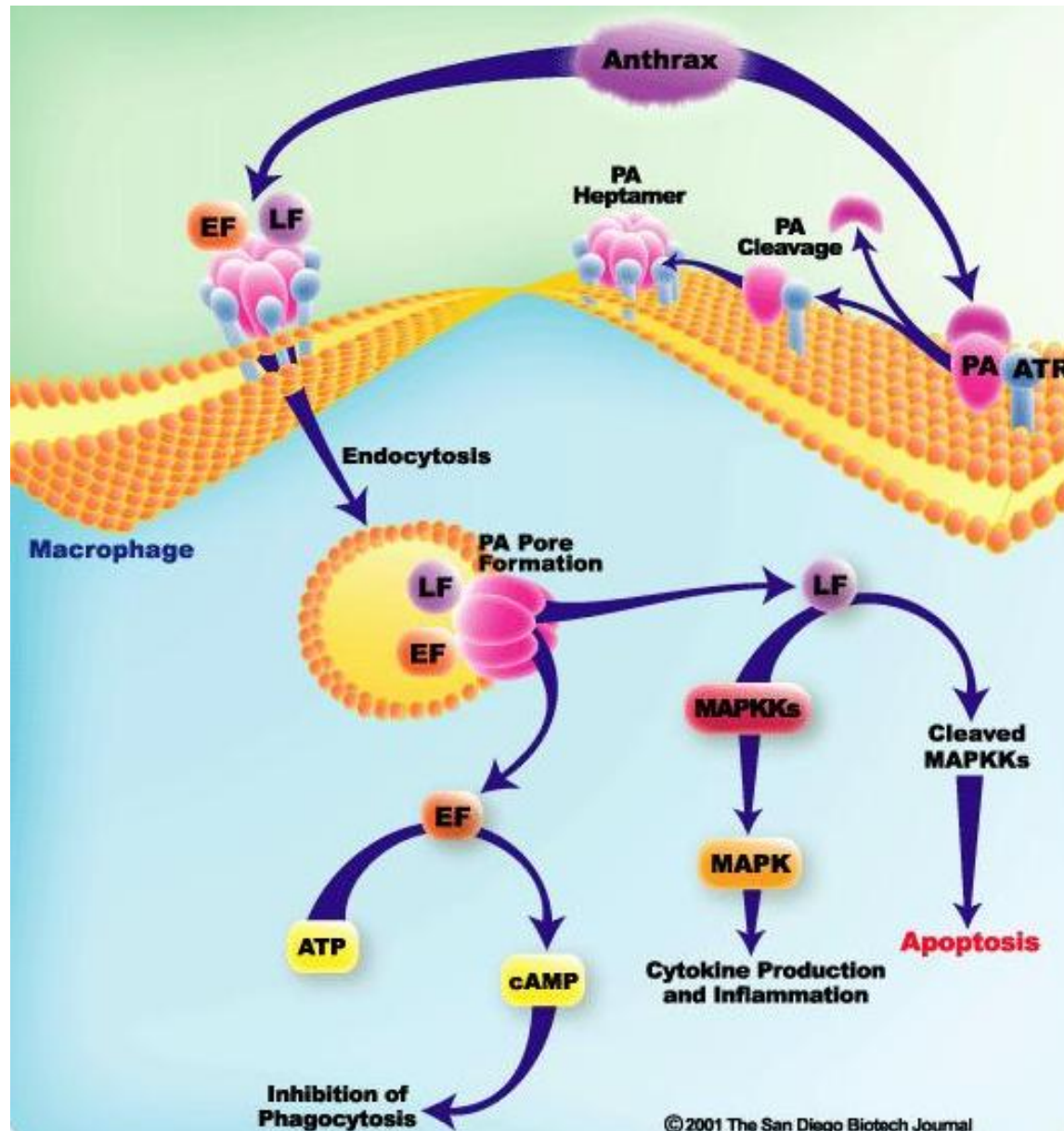


mitogen-activated protein kinase cascades, an ubiquitous signaling “submodule” in eukaryotes, involved in proliferation, differentiation, movement, death (receptor tyrosine kinase as input; control of transcription factors)

Different MAPK Cascades



Even: Role in Anthrax

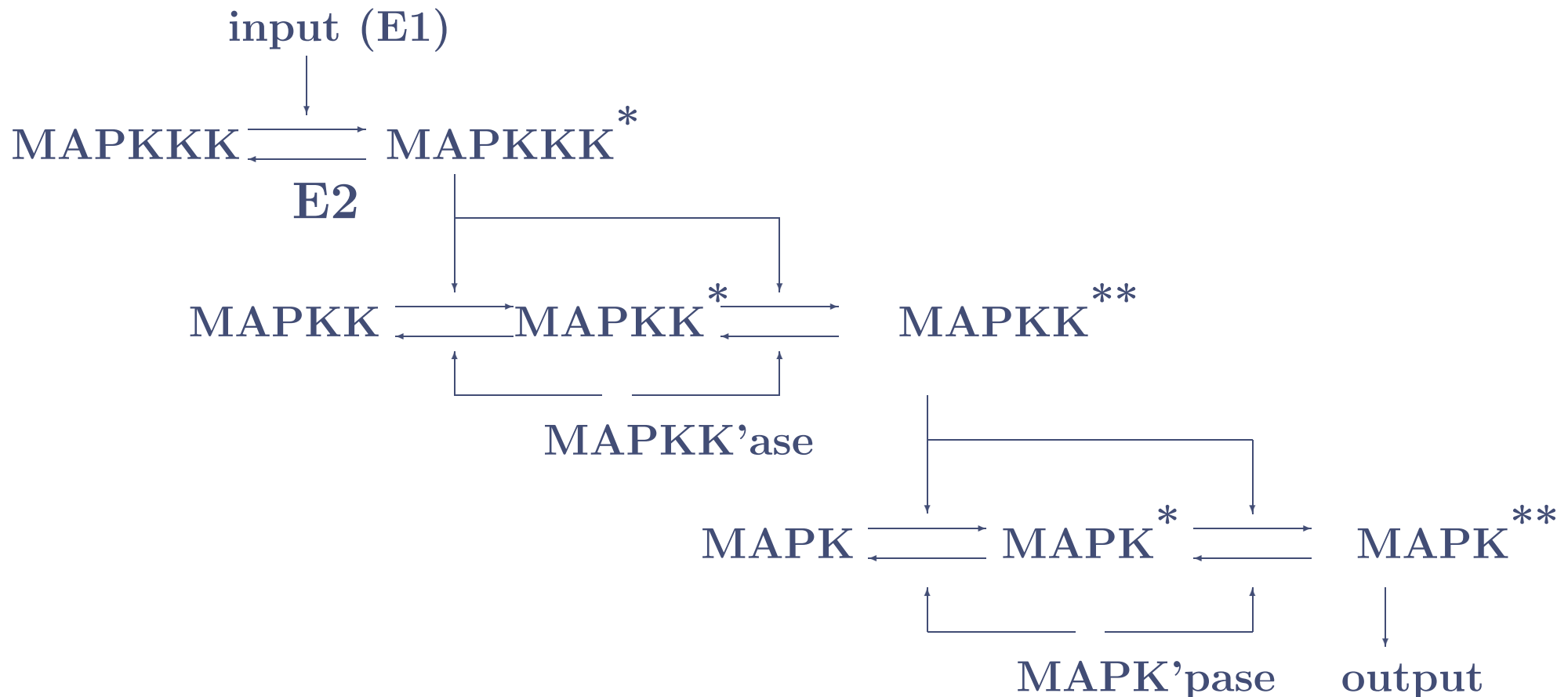


Lethal Factor (LF) is a toxin secreted by *Bacillus anthracis*. It is a highly specific protease that cleaves members of a MAPKK near their amino-termini, removing the docking sequence for the down-stream cognate MAP kinase

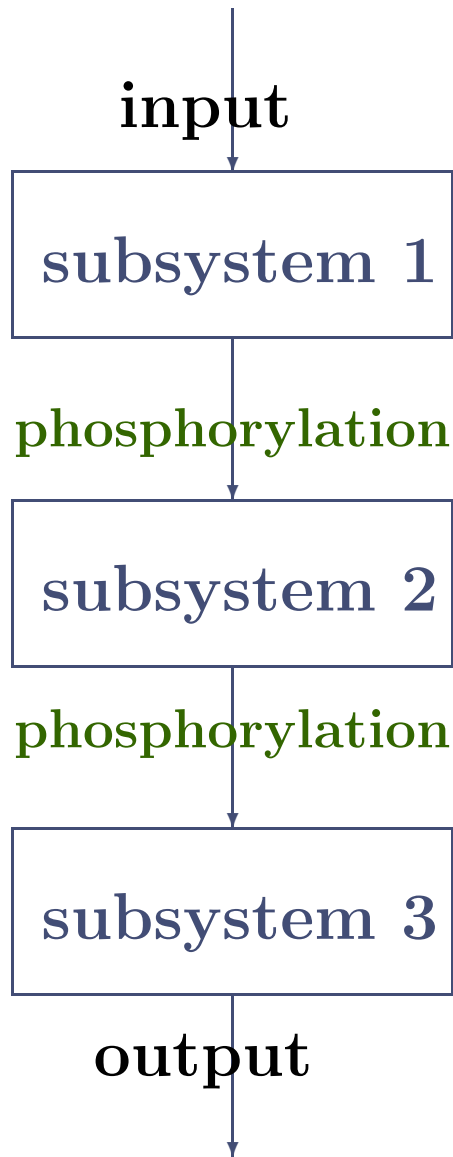
Common Structure

phosphorylated form $\rightsquigarrow$ tyrosine phosphorylation of next
reverse degradation by phosphatase

Huang-Ferrell PNAS '96 model:



Some Results



basic mechanism (with variations):
cascade of three subsystems,
each foliated into invariant hypersurfaces

each level is a *monotone system*
(Angeli & EDS, CDC 2002):

$\exists$ partial order s.t. flow order preserving
($\equiv$ all linearizations are positive systems)
 $\therefore$ monotone I/O of each, $\therefore$ cascade too

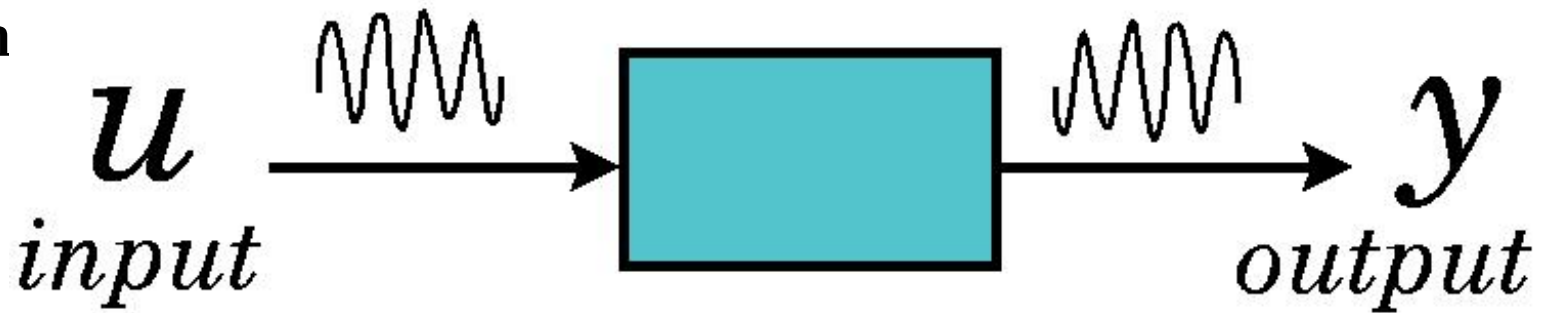
& $\exists$ well-defined “I/O characteristic” φ :
 $u(t) \equiv \mu \Rightarrow y(t) \rightarrow \varphi(\mu)$ as $t \rightarrow +\infty$
(typically assumed true, w/o proof!)

moreover, *Cauchy gain* well-defined:

$\forall$ signal ω defined on $[0, +\infty)$ (values in Banach space)
define **asymptotic amplitude**:

$$\|\omega\|_{\text{aa}} := \limsup_{s, t \rightarrow \infty} \|\omega(t) - \omega(s)\| \in [0, \infty]$$

system



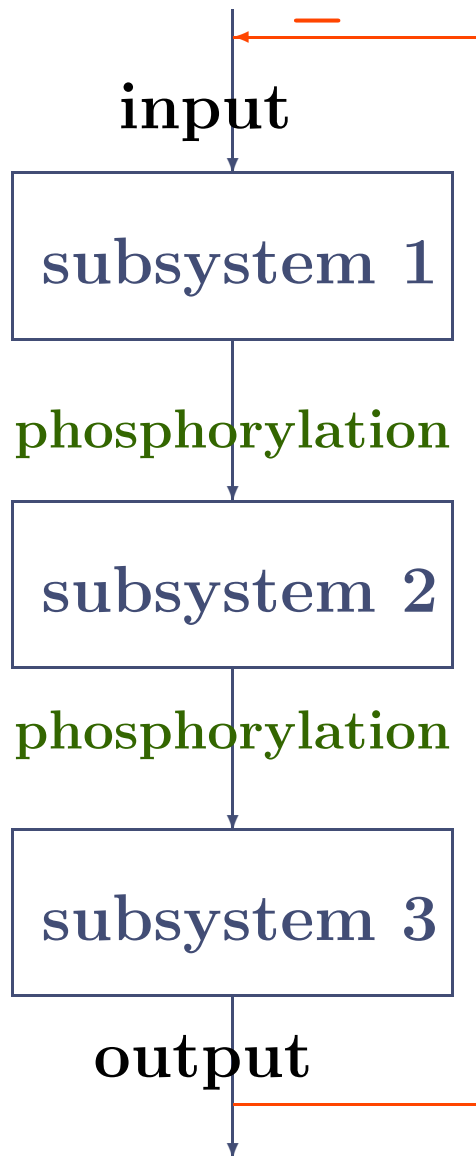
admits Cauchy gain $\gamma \in \mathcal{K}_\infty$ if $\|y\|_{\text{aa}} \leq \gamma(\|u\|_{\text{aa}}) \quad \forall u \rightsquigarrow y$

since $\|\omega\|_{\text{aa}} = 0 \iff \exists \omega^\infty := \lim_{t \rightarrow \infty} \omega(t)$ (Cauchy!),

$\therefore$ “converging input $\Rightarrow$ converging output”

introduced in (SCL’02) in order to deal with this question:

Oscillations



Kholodenko, Europ. J. Biochemistry 2000
& Shvartsman/Wiley/Lauffenburger 2000:

negative feedback (to turn-off response??)

(inhibitory phosph. of SOS = “Son of Sevenless” by p42/p44 MAPK = ERK)

may induce oscillations (Hopf bifurcation)
if too strong ... how strong is too strong?

obvious approach: a small-gain theorem

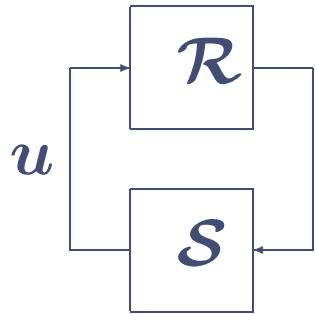
ask “loop gain  less than 1”

but gain in what sense?

*in biology, the equilibrium location
often depends on the feedback gain!*

e.g.: $\dot{x} = -x + \frac{4}{3+x} \Rightarrow$ equilibrium moves from $x = 0$ to 1

Trivial SGT for Cauchy Gains:



closed-loop signals =
intersection of graphs: $\mathcal{R} \cap \mathcal{S}^{-1}$

if $\mathcal{R}$ and $\mathcal{S}$ have Cauchy gains γ_1 and γ_2 respectively, and

$$\gamma_1(\gamma_2(r)) < r \quad \forall r > 0$$

then all bounded closed loop signals converge to equilibria:

$$\|y\|_{aa} \leq \gamma_1(\gamma_2(\|y\|_{aa})) < \|y\|_{aa} \Rightarrow \|y\|_{aa}=0 \Rightarrow \|u\|_{aa}=0$$

also: incremental gain ($\sim$ Lipschitz) $\Rightarrow$ uniqueness

with these concepts, all work goes into estimating gains

gain of cascade composes; delays OK; Lyapunov; $\text{diam}(\Omega^+)$

Application to MAPK Model

e.g. single-phosphorylation, eqs for activated form
nondimensionalize to total “1”, state space $[0, 1]^3 \subset \mathbb{R}^3$

$$\begin{aligned}\dot{x}_1 &= -\frac{b_1 x_1}{c_1 + x_1} + \frac{\mu}{1 + k x_3} \frac{d_1(1 - x_1)}{e_1 + (1 - x_1)} \\ \dot{x}_2 &= -\frac{b_2 x_2}{c_2 + x_2} + x_1 \frac{d_2(1 - x_2)}{e_2 + (1 - x_2)} \\ \dot{x}_3 &= -\frac{b_3 x_3}{c_3 + x_3} + x_2 \frac{d_3(1 - x_3)}{e_3 + (1 - x_3)}\end{aligned}$$

μ = constant input, k = strength of inhibition

with coefs from Shvartsman et. al., MIT Report'2000

$b_1 = c_1 = e_1 = b_2 = 0.1, c_2 = e_2 = c_3 = e_3 = 0.01, b_3 = 0.5, d_1 = d_2 = d_3 = 1$

bifurcation analysis $\Rightarrow$ oscillation onset @ $k = 5.2, \mu = 0.3$

SCL'02 Cauchy SGT: OK global stability if $k < 3.918$
(even if $\exists$ arbitrary delays)

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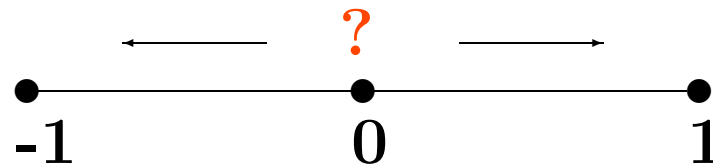
■ *discussion*

Necessity of Switching

just like “IMP $\Rightarrow$ embedded signal generator”,
 $\exists$ deep theory characterizing necessity and/or sufficiency
of discontinuities and hysteresis in feedback loops

trivial example, to illustrate ideas:

point in interval $[-1, 1]$, want to drive to boundary
may move left/right at any speed (but stay put if there)

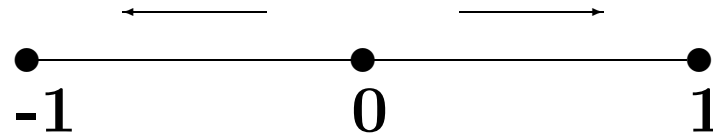


obviously: decision rule must be discontinuous
e.g. “go right if $x > 0$; left otherwise”

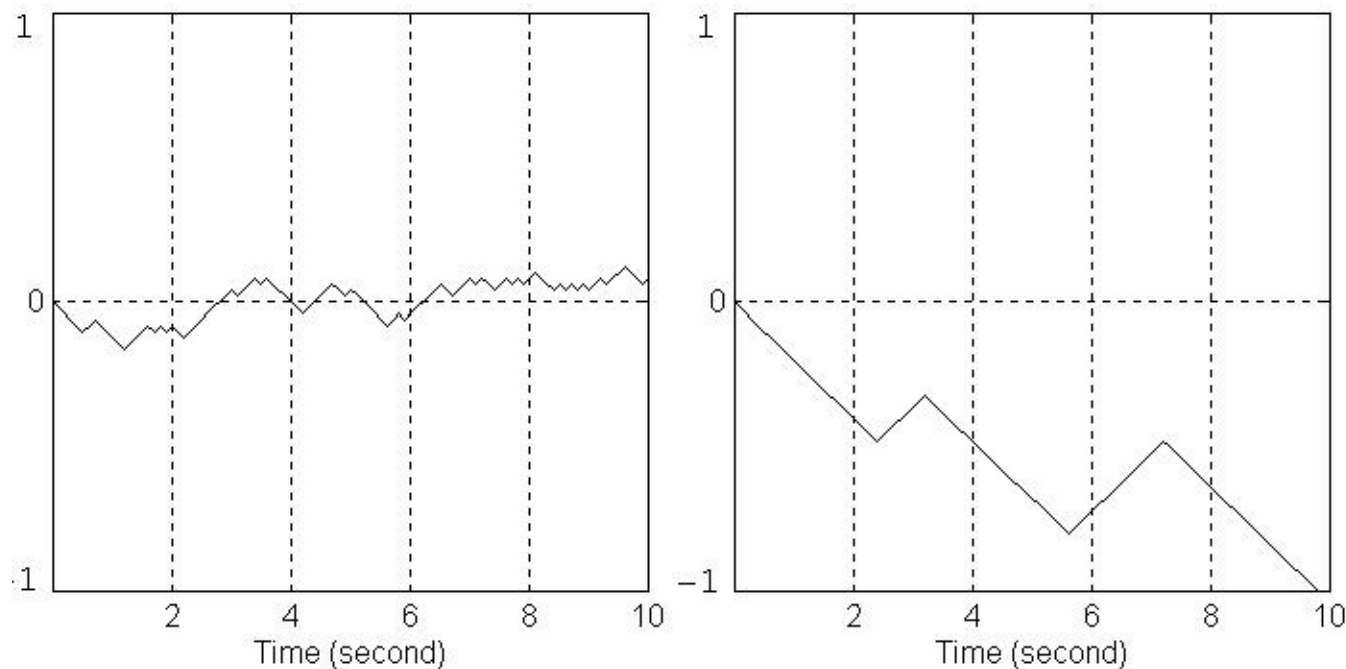
general theory: drive system to target configuration
need for switches not obvious: topological obstructions,
or “virtual obstacles” due to nonholonomy (e.g. non-slip)

Necessity of Memory

now assume there is (small) noise in observations



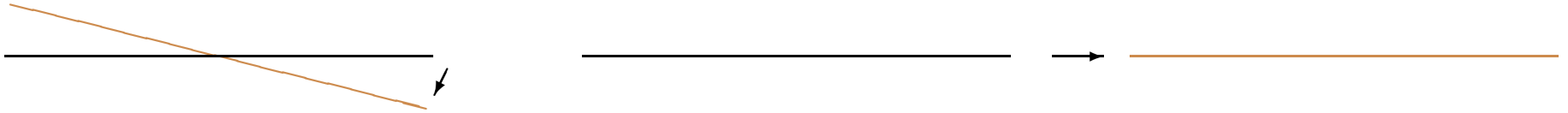
may have chattering due to noise (worst case: stuck)
solution: *hold* decision for small time; don't observe



cannot do w/o memory in feedback
— theorems for very general systems, but first another e.g.

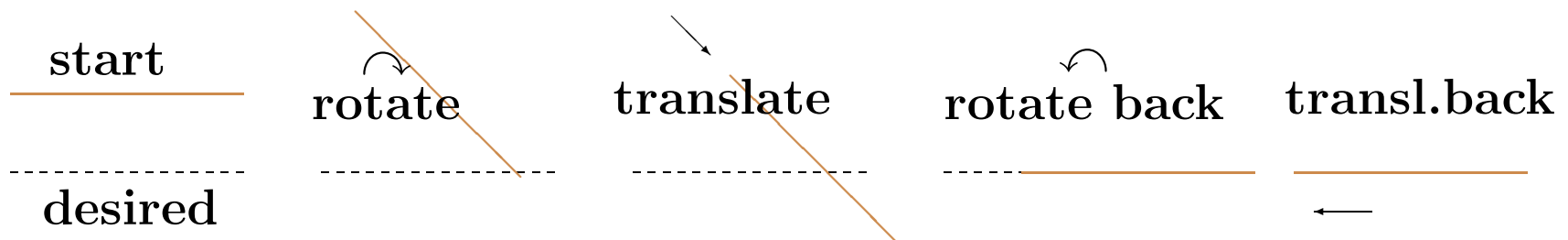
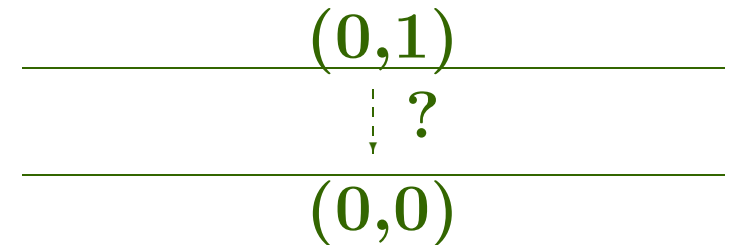
A More Interesting Example

stick can rotate around its center, or displace along its axis



“turn and drive” (or “tumble and run”?) system
(3-dim system: orientation, coords of the center)

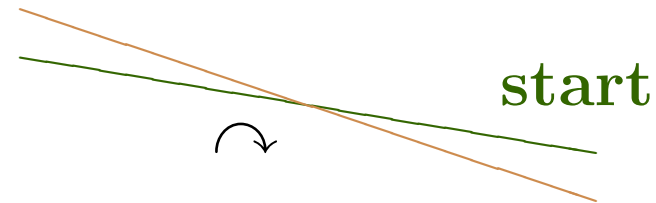
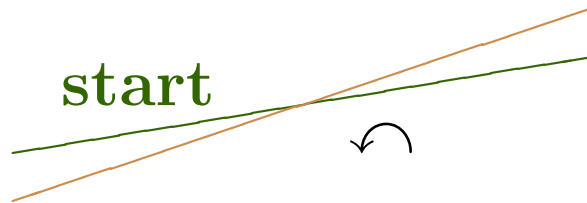
“no parallel translations” means
going $x_0 = (0, 0, 1) \rightsquigarrow (0, 0, 0)$
requires “parallel-parking” motion



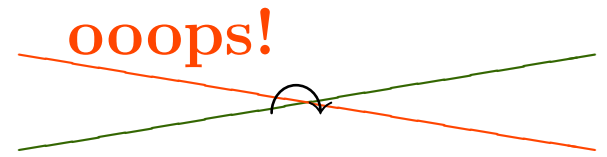
i.e., Lie (commutator) motions are required $(r \ t \ r^{-1} \ t^{-1})$
 $\Rightarrow$ discontinuities required (degree theory - cf. MCT)

Noise $\Rightarrow$ Need “Memory”

suppose rule: if “\” turn CW, if “/” turn CCW

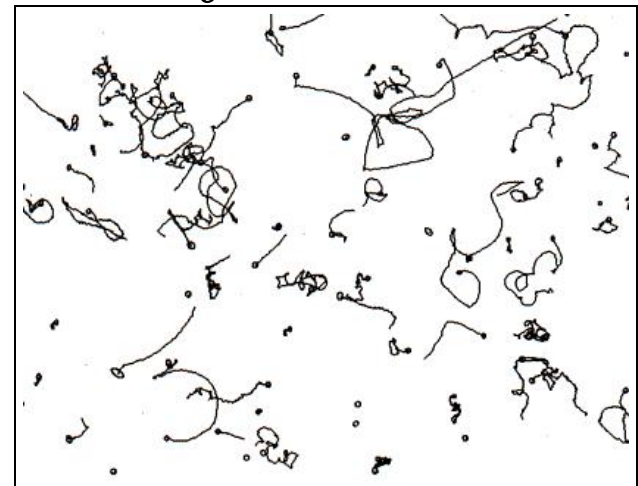


starting from “/” measurement noise may fool controller into believing opposite, so rotation in the wrong direction



“malicious” disturbance might consistently fool controller

chattering again avoidable
by “stay the course” method
...tumbles and “long runs”
consistent with general theory



General Theorem

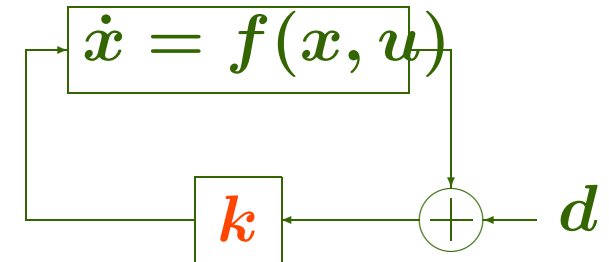
quite general ODE's $\dot{x} = f(x, u)$, just loc. Lipschitz

objective: find a feedback law k

such that closed-loop system has target as globally asymptotically stable point

even if $\exists$ measurement errors (noise) $d(\cdot)$

general theorem combines many ingredients:



Hermes 1967 (measurement errors),

Hájek 1979 (relation Hermes, Krasovskii, Filippov),

Sussmann 1979 (examples of the necessity of discontinuous feedback),

Sussmann-EDS 1980 (memory sufficient, 1-dim),

EDS 1981 ($\mathcal{C}^0$ clf's),

Artstein 1983 ($\mathcal{C}^1$ clf's),

Brockett 1983 (Lie conditions, Krasnosel'ski's theo),

Coron 1992 (time-varying, no drift),

Lin-EDS-Wang 1993 (converse Lyapunov),

Coron-Rosier 1994 (converse, measurable selections),

Clarke-Ledyaev-EDS-Subbotin 1997 (general existence theorem),

Ledyaev-EDS 1998 (measurement error obstructions),

EDS 1999 (clocks sufficient)

Theorems

- stabilization always possible
under obvious “asymptotic controllability” conditions
- characterize when discontinuities required (via V)
- with noise, in general $\nexists$ memoryless controller
(even non- $\mathcal{C}^0$) which prevents chattering
- but OK if using memory or “hysteresis” strategy

Clarke-Ledyaev-EDS-Subbotin 1997 *IEEE Trans. Automatic Control*

Ledyaev-EDS 1998 *Nonlinear Analysis*

EDS 1999 *Control, Optimization, Calculus of Variations*

intuitive idea: use same control, not re-estimating state for
minimal period $T = T(\text{system eqs, Lipschitz constants, etc})$

compare, but not quite same: deadzones, thermostat

Outline: Impossibility Proof

*

the impossibility result is based on proving that, if some controller k stabilizes under small measurement errors,

$$\dot{x} = f(x, k(x + \epsilon))$$

then there must also be a **continuous** stabilizer, and hence the usual necessary conditions apply

this is done by showing that an associated usc differential inclusion is strongly stable,

$$\dot{x} \in \bigcap_{\delta > 0} \overline{\text{co}} f(x, k(x + \delta B))$$

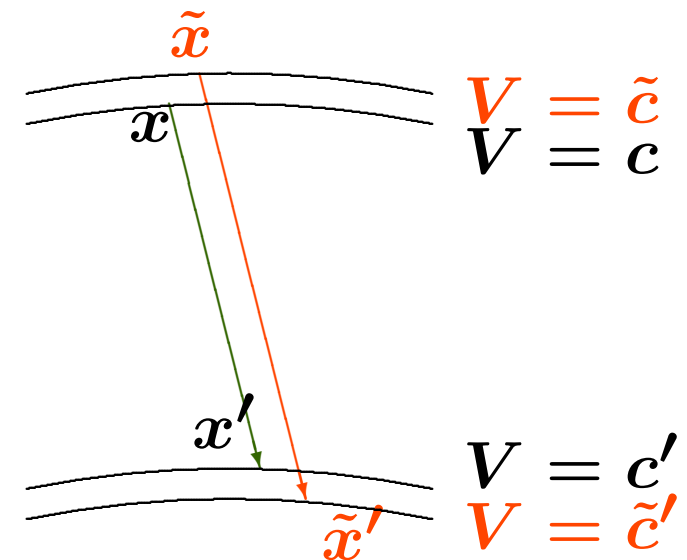
so converse theorem provides smooth Lyapunov function V , which is then a smooth cLf for the original system and thus a continuous stabilizer exists

Outline: “Stay the Course” Construction

one first shows, for any asymptotically controllable system, the existence of a **continuous** control-Lyapunov function V , as value function of appropriate optimal control problem

then V is regularized using a Iosida-Moreau inf-convolution and the feedback k is obtained by solving an associated (viscosity) Hamilton-Jacobi partial differential inequality

let $\tilde{x} := x + \varepsilon \approx x$, use $u := k(\tilde{x})$ for time long enough s.t. $\tilde{c}' \ll \tilde{c}$
 controller thinks $\rightsquigarrow \tilde{x}'$ but really $\rightsquigarrow x'$
 continuity on initial conds $\Rightarrow c' \approx \tilde{c}'$
 $\therefore c' \approx \tilde{c}' \ll \tilde{c} \approx c$ i.e. V decreases



technical issues include giving precise meaning to “**solution**” of closed-loop system $\dot{x} = f(x, k(x))$ when k is discontinuous — done in ways reminiscent of Itô solutions of stochastic ode’s

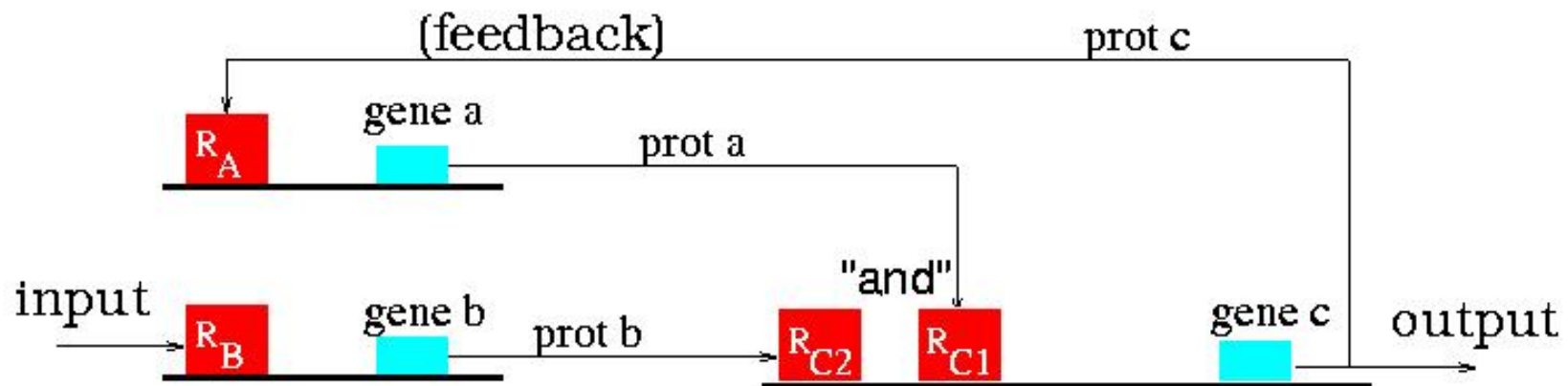
Hybrid Systems

saw discontinuities in feedback laws may be unavoidable

indeed, control mechanisms in cells involve switching
(e.g. bind activator or repressor regulatory regions)

may model switches by bistable differential equations
(faster time scale than rest of system)

...or hybrid models which combine discrete and continuous components, and combinatorial logic to implement Boolean operations on the discrete components

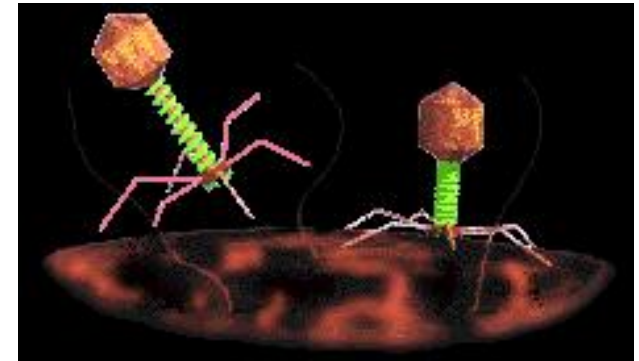


λ phage control circuit

bacteriophage viruses

(Twort & d'Hérelle, 1915/7)

λ infects *E.coli*



T4

dormant (**lysogeny**) until senses lethal danger to its host
e.g. UV radiation $\Rightarrow$ “abandon ship” (**lysis**) & infect others

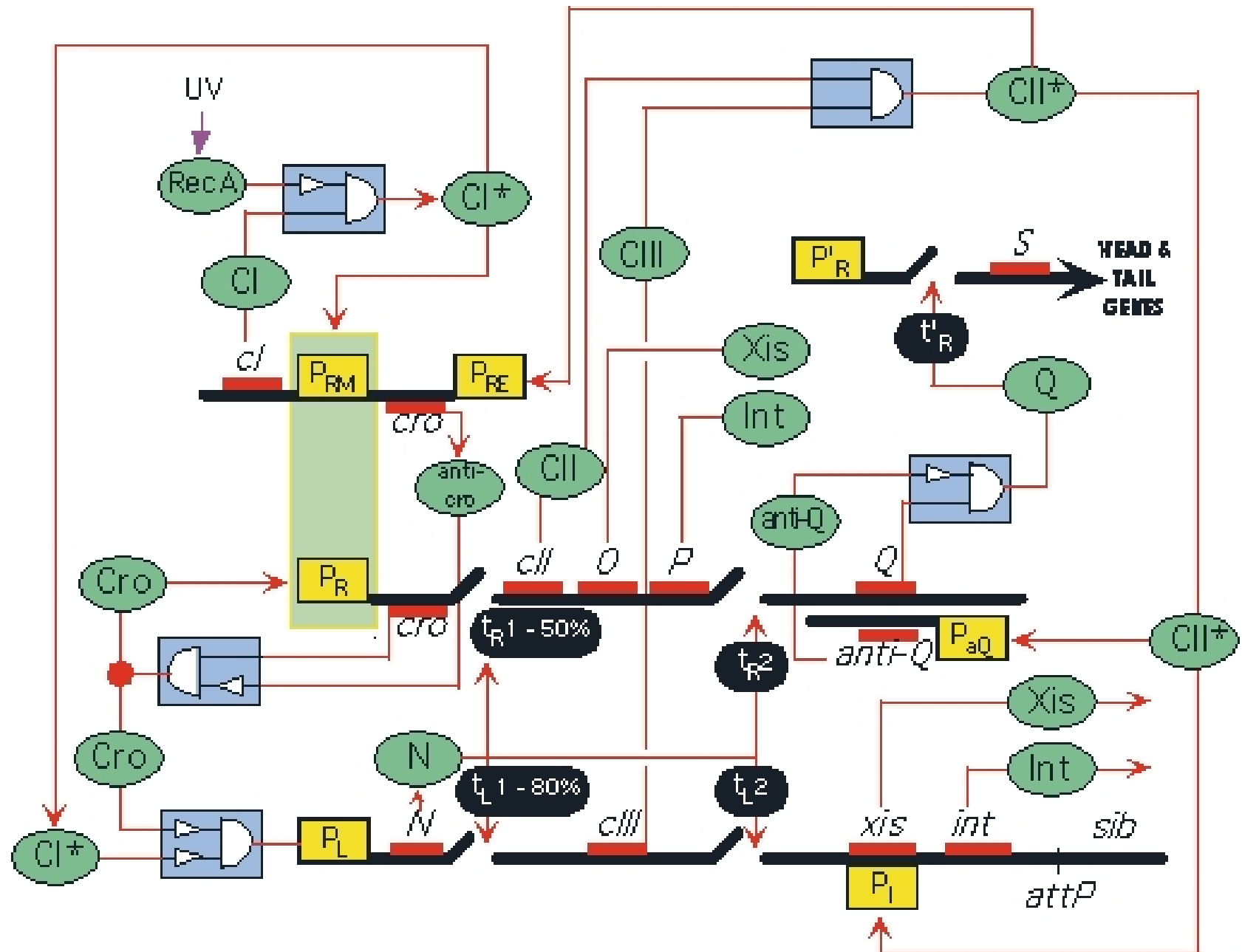
binary lysogeny/lysis state decided by competition
for binding sites between **Cro** and **λ repressor** proteins

UV $\Rightarrow$ RecA (repair) protein activated $\Rightarrow$ degrade repressor;
otherwise, λ keeps control

1995 paper gave biologically-based, hybrid simulation
model of the lysogeny/lysis decision subsystem in λ
goals: complete wiring diagrams for larger systems
build artificial circuits for therapeutic purposes

McAdams and Shapiro, Circuit Simulation of Genetic Networks, *Science* 1995

λ phage control circuit – goal: other wiring diagrams & *design* [ident/realiz]



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Some Remarks: ODE's

chemical concentrations $\in \mathbb{R}$: population avg., expectations

contrast mech systems: steady state $\neq$ “lack of activity”
rather: balance between creation and absorption rates
(or equilibrium probability distributions)

but low molecule #'s (10's-100's), so mean approximations
valid?

active research: diffusion approximations for “internal noise”
 $\rightsquigarrow$ stochastic PDE's, understanding of stability consequences

spatial homogeneity in concentrations (ode vs pde's):
for small (intracellular) volumes, diffusion plays a major role
but also localization: cell membrane, nucleus, ribosomes
 $\rightsquigarrow$ compartmental systems, model transport mechanisms

On Models and Theory

avoid “pseudo-exactness”!

in engineering (including bioengineering, e.g. heart control, prostheses), it makes sense to develop high-precision models

but in molecular biology, relevant values (rate constants, concentrations) may vary over orders of magnitude

∴ interest in results robust to large parameter variations
– conclusions not meaningful if depending on exact values

structure should provide **robustness** of desired behavior

e.g. Von Dassow et.al., Nature'2000, segment polarity network in *Drosophila* embryos: topology encodes function, even under huge changes in constants and even functional forms of connections

[cellular output & organism-level quantities precise: body temperature and pressure, glucose concentrations, frequencies of circadian clocks]

e.g. optimal control: good algorithms not as useful as
general theory - *types* of solutions: bang-bang, singular?

role of optimality: evolution produced “*locally* optimal” solutions?

inverse optimal design theory applicable?

Conclusions

feedback control theory deals, in principle, with many issues of interest in postgenomic (“systems”) molecular biology

one may expect that many **principles** of feedback theory will be of great use in **guiding experimental work**

while parts of the theory (e.g. identification, observers) may play a role in experimental design and instrumentation

in addition, extremely interesting research areas in mathematics arise when looking at these problems

e.g. from my own experience: new IMP's, Cauchy gains, monotone systems, stability of chemical nets

<http://www.math.rutgers.edu/~sontag>